

International Journal of Human–Computer Interaction

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/hihc20

FactExplorer: Fact Embedding-Based Exploratory Data Analysis for Tabular Data

Qi Jiang, Guodao Sun, Yue Dong, Lvhan Pan, Baofeng Chang, Li Jiang, Haoran Liang & Ronghua Liang

To cite this article: Qi Jiang, Guodao Sun, Yue Dong, Lvhan Pan, Baofeng Chang, Li Jiang, Haoran Liang & Ronghua Liang (12 Feb 2025): FactExplorer: Fact Embedding-Based Exploratory Data Analysis for Tabular Data, International Journal of Human–Computer Interaction, DOI: 10.1080/10447318.2025.2459172

To link to this article: https://doi.org/10.1080/10447318.2025.2459172

View supplementary material

Published online: 12 Feb 2025.

Submit your article to this journal

Article views: 42

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=hihc20

INTERNATIONAL JOURNAL OF HUMAN–COMPUTER INTERACTION
https://doi.org/10.1080/10447318.2025.2459172

Taylor & Francis
Taylor & Francis Group

Check for updates

FactExplorer: Fact Embedding-Based Exploratory Data Analysis for Tabular Data

Qi Jiang, Guodao Sun, Yue Dong, Lvhan Pan, Baofeng Chang, Li Jiang, Haoran Liang, and Ronghua Liang

College of Computer Science and Technology, Zhejiang University of Technology, Hangzhou, China

ABSTRACT

Despite exploratory data analysis (EDA) is a powerful approach for uncovering insights from unfamiliar datasets, existing EDA tools face challenges in assisting users to assess the progress of exploration and synthesize coherent insights from isolated findings. To address these challenges, we present FactExplorer, a novel fact-based EDA system that shifts the analysis focus from raw data to data facts. FactExplorer employs a hybrid logical-visual representation, providing users with a comprehensive overview of all potential facts at the outset of their exploration. Moreover, FactExplorer introduces fact-mining techniques, including topic-based drill-down and transition path search capabilities. These features facilitate in-depth analysis of facts and enhance the understanding of interconnections between specific facts. Finally, we present a usage scenario and conduct a user study to assess the effectiveness of FactExplorer. The results indicate that FactExplorer facilitates the understanding of isolated findings and enables users to steer a thorough and effective EDA.

KEYWORDS

Data insight; data embedding; exploratory data analysis; human-computer interaction

1. Introduction

Exploratory data analysis (EDA) serves as a fundamental and powerful approach for understanding data and deriving insights (Jankun-Kelly et al., 2007). Most existing EDA methods are data-based, and their visualization pipeline (Figure 1-top) generally anchors on raw data and includes steps such as data transformation and visual mapping (Keim et al., 2010). These steps are iteratively refined through user feedback, allowing for progressively deeper exploration. Recently, some studies have sought to streamline the process by automating data transformation and visual mapping steps (Cui et al., 2019; Hu et al., 2019; Jiang et al., 2024; Mackinlay et al., 2007; Vartak et al., 2015; Zhu et al., 2020), thereby enabling users to identify meaningful patterns more efficiently.

Despite the advances in these data-based EDA approaches, they still face certain limitations. Firstly, most of these methods follow an incremental knowledge discovery process, where users gradually accumulate an understanding of the data through iterative data analysis operations, either guided by system recommendations or based on personal experience and intuition (Jebb et al., 2017; Wongsuphasawat et al., 2019). However, this incremental approach often lacks a global perspective. As a result, users may easily fall into local traps, where they become overly focused on specific data characteristics or patterns while overlooking other potentially important information. Moreover, tracking the progress of exploration becomes challenging, making it difficult for users to clearly discern which areas have been examined and which require further investigation. Secondly, although EDA research has introduced a range of flexible and effective interaction techniques (e.g., comparing (Law et al., 2019), drill-down (Joglekar et al., 2019; Law et al., 2018), and filtering (Amar et al., 2005)) to encourage exploration and facilitate discovery. However, the findings obtained may sometimes appear isolated or unrelated, particularly when they involve different attributes or are represented through diverse visual encodings. Clarifying the connections between these isolated findings can help users develop a comprehensive understanding, but current EDA tools still fall short in this regard.

To bridge these gaps, we introduce a novel fact-based EDA framework (Figure 1-bottom), which shifts the analysis focus from raw data to data fact. This framework prioritizes the proactive extraction of potential facts from the data, followed by a higher-level exploration centered around these facts. In the visualization community, data fact has been described in various ways (Lu et al., 2021; Shi et al., 2021; Srinivasan et al., 2018; Wang et al., 2020). This paper defines it as a piece of information extracted from data and presented as a visualization chart. It results from a focused examination of an input dataset by applying a specific analytical task. Each data fact is composed of the following key components: its type (*Distribution*, *Correlation*, *Trend*, *Proportion*, and *Aggregation*), each with its analytical expectations and goals (e.g., a *Correlation* fact aims to find patterns or dependencies between variables); data subset, which refers to the particular collection of data items used for fact generation; and observation dimensions, which involve the

CONTACT Guodao Sun guodao@zjut.edu.cn College of Computer Science and Technology, Zhejiang University of Technology, Hangzhou, China
Supplemental data for this article can be accessed online at https://doi.org/10.1080/10447318.2025.2459172.
© 2025 Taylor & Francis Group, LLC

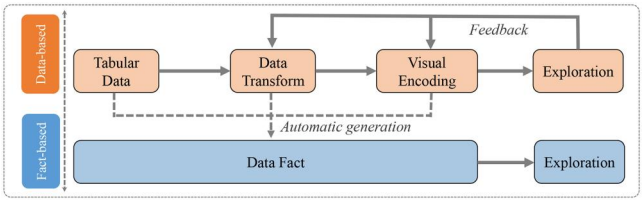


Figure 1. The illustration of data-based and fact-based EDA, inspired by Keim et al. (2010). The former focuses on raw data and employs a bottom-up and iterative analytical process to delve into the intrinsic patterns within the data. The latter emphasizes the proactive extraction of potential data facts from the data, followed by a top-down and higher-level exploration centered around these facts.

specific attributes within this data subset that are analyzed. Furthermore, each fact is assigned an interestingness score based on its significance (statistical characteristics) and impact (the amount of raw data involved in fact generation).

Our analytical framework consists of three modules: fact extraction, fact embedding, and fact exploration. Given a tabular dataset, the system automatically extracts potential data facts. These extracted facts are then vectorized using a hybrid logic-visual embedding method for a more comprehensive representation. The visual representation focuses on capturing the visual structure of fact presentations employing a Grammar Variational Autoencoder (GVAE) (Kusner et al., 2017; Zhao et al., 2022). This encoding process aims to group visually similar facts (in terms of their graphical layout and appearance) closer together in the semantic space. The logical representation seeks to model the data relationships between facts by evaluating four predefined logical relationships (*Parallelism*, *Granularity*, *Temporality*, and *Contrast*). These relationships indicate how facts are interconnected or related in a meaningful way, such as *Granularity*, which reflects differences in data detail where one fact may encompass or be a more granular version of another. This logical similarity is then combined with the visual similarity derived from the features generated by the GVAE through a weighted summation, yielding a comprehensive embedding vector. Comparative experiments of different feature fusion methods were conducted to trade off their pros and cons. Beyond fact embedding, the framework offers a topic-based drill-down function for querying other facts related to an original fact of interest. For any two facts specified by users, we enable producing a transition path between them through a series of transformation operations, where the generated intermediate facts from this process aid in elucidating the connection between given facts.

The abovementioned ideas are implemented in a fact-based tabular data analysis system, FactExplorer. We demonstrate a usage scenario and conduct a user study using real-world datasets to evaluate FactExplorer's usability and effectiveness. Moreover, FactExplorer is also available as an open-source project[1], contributing to the ongoing research in EDA. The primary contributions of this work are as follows:

- A novel hybrid logical-visual fact embedding approach designed to provide users with a comprehensive overview of data facts, and a comparative study of different feature fusion methods to trade off their pros and cons.
- A suite of interactive exploration techniques: topic-based drill-down and transition path generation. The former enables a systematic and effective in-depth analysis of data facts, while the latter supports users in understanding how specified facts are related.
- A fact-based data analysis framework is implemented in a runnable system named FactExplorer, and its effectiveness is thoroughly evaluated through a usage scenario and a user study using real-world datasets.

2. Related work

In this section, we elaborate on the recent studies that are most relevant to this work, including data embedding and exploratory data analysis.

2.1. Data embedding

Data embedding is a process that involves transforming high-dimensional data into a lower-dimensional feature space to align semantically similar data pairs closely. Dimensionality reduction (DR) is a widely recognized method for embedding and is precious for compressing multi-dimensional data (Sun et al., 2022). DR methods are generally divided into linear techniques, exemplified by principal component analysis (PCA) (Abdi & Williams, 2010), and non-linear techniques, including multi-dimensional scaling (MDS) (Borg & Groenen, 2005), uniform manifold approximation and projection (UMAP) (McInnes et al., 2018), and t-distributed stochastic neighbor embedding (t-SNE) (Van der Maaten & Hinton, 2008). DR techniques reduce computational overhead and storage needs while maintaining the structure inherent to the original data.

Beyond its utility in DR, data embedding serves as a fundamental preprocessing step in various complex computational fields like natural language processing (NLP) (Devlin et al., 2019; Mikolov et al., 2013), computer vision (Cheddad et al., 2010), and graph analysis (Cai et al., 2018). Recently, the visualization community has also shown a growing interest in visualization similarity and chart embedding, aiming at facilitating downstream tasks such as visualization recommendation (Zhu et al., 2020) and chart summarization (Xu et al., 2018; Zhao et al., 2022). Early explorations primarily used graph models to measure the similarity between images or between charts. For instance, the Image Graphs model (Ma, 1999) treats images as graph nodes linked by edges representing parameters, employing various feature channels to render multiple images. The P-Set Model (Jankun-Kelly et al., 2007) replaces parameters with exploration processes that utilize various visualization techniques. Following this, Hullman et al. (2013) conceptualized the visualization state space through directed graphs where edges symbolize transitions between charts. GraphScape (Kim et al., 2017) takes this a step further by presenting cost functions to model visualization similarity. Beyond graph-based approaches, integrating machine

learning into visualization embedding represents an emerging trend (Demiralp et al., 2014). For instance, GVAE (Kusner et al., 2017) was developed to generate coherent latent representations of discrete data. ChartSeer (Zhao et al., 2022) extends GVAE's methodology by embedding Vega-Lite (Satyanarayan et al., 2017) formatted charts into latent vectors.

Current chart embedding techniques mainly rely on graphical features, which limits their ability to comprehensively describe charts. In contrast, hybrid embedding approaches that represent charts from multiple perspectives contribute to a comprehensive representation (Xie et al., 2021; Xu et al., 2018). In this work, we propose a novel hybrid embedding model that considers both the inherent visual features of each fact and the logical relationships among various facts. Firstly, to our knowledge, our model is the first to incorporate logical relationships between facts in a visualization representation, which is essential for forming narrative structure and enhancing data analysis (Hullman et al., 2013; Shi et al., 2021). Secondly, the model is human-centered, with its dual considerations ensuring that the output results better align with users' analytical behavior models. Specifically, in the context of EDA, users typically focus their further examination and exploration on charts that have similar visual encodings or are related through data lineage to the current visualization (Wongsuphasawat et al., 2016, 2017). Consequently, the surrounding data facts for each data point in the overview should reflect this relevance, making it easy and efficient for users to identify contextual information. Moreover, our model is interactive and extensible, allowing users ample flexibility to customize feature weights and add semantic features.

2.2. Exploratory data analysis

The concept of EDA originates from John Tukey's early work (Tukey, 1977), whose primary objective is to understand data patterns, uncover insights, and formulate and validate hypotheses. EDA encompasses a series of incremental steps (i.e., interactions) tailored to achieve specific goals, which can range from well-defined tasks (e.g., exploring existing hypotheses or intuitions) to relatively vague aims (e.g., finding interesting something in the data without predefined expectations).

Throughout the EDA process, analysts are engaged in a continuous decision-making endeavor. A recent research trend involves integrating guidance-based techniques to facilitate this process. For instance, Show Me (Mackinlay et al., 2007) automatically suggests visual encoding designs based on the statistical features of the selected data. Similarly, Voyager (Wongsuphasawat et al., 2016) and Voyager 2 (Wongsuphasawat et al., 2017) employ heuristic approaches to recommend alternative visualization designs, aiming to inspire visual analysis exploration. Foresight (Demiralp et al., 2017; Demiralp et al., 2017) introduces a rule-based approach to extract insights, allowing users to query neighboring insights further. Later, QuickInsights (Ding et al., 2019) standardizes the format of various insights and provides a method for evaluating the interestingness of insights according to data characteristics. In addition, guidance-based exploration is also introduced to other analytical scenarios, e.g., Clustrophile 2 (Cavallo & Demiralp, 2018) guides users in conducting cluster-based exploratory analysis and incorporates user feedback to enhance the guidance and facilitate the interpretation of clusters.

More recently, several researchers recognized the potential of guidance in EDA, they also identified certain limitations (Ceneda et al., 2017; Sperrle et al., 2023; Stoiber et al., 2022). A primary challenge is the balance between providing helpful guidance and preventing users from becoming overly reliant on such assistance, which may impede independent exploration (Li et al., 2021; Vartak et al., 2017). In this work, we attempt to avoid this pitfall by adopting an "overview + exploration" approach. Instead of limiting users to a narrow set of system-recommended perspectives, we provide an overview of the entire potential insight space. On the one hand, we alleviate the burden of manual analysis operations through automated fact extraction. On the other hand, by presenting a fact overview, we open up the "black box" of the pattern space that is hidden in traditional EDA tools (Pérez-Messina et al., 2022), providing ample room for users to actively explore.

In summary, departing from the existing methods, we introduce a novel top-down EDA paradigm that first pre-extracts data facts, then provides a fact overview, followed by fact-based exploration within this context. While we draw upon existing data fact extraction methods from domains such as automated data insights (Lu et al., 2021; Shi et al., 2021; Wang et al., 2020), our application and integration of these methods into EDA provide a fresh perspective to understanding complex datasets. Moreover, our approach also introduces multiple analytical functionalities derived from these data facts, including revealing connections between user-specified facts and querying facts related to a given topic. These capabilities extend beyond traditional EDA tools, facilitating the user's discovery of data patterns and understanding of associations between facts. Overall, our approach complements the existing landscape of EDA and enriches the toolbox available for analysts seeking to discover meaningful insights from their data.

3. FactExplorer

In this section, we elaborate on the research challenges and formulate the design requirements. Moreover, the system framework, user interface, and interaction of FactExplorer are also described.

3.1. Challenges

Our research aims to design and implement a novel EDA system to assist data analysts in more comprehensively exploring data and fostering deeper insights. However, existing visualization methods face the following challenges in achieving this goal:

C1 Lack of a holistic overview of data facts within the data. Most current EDA methods lack the ability to explicitly reveal the data facts landscape from the outset. Users are often required to iteratively explore the data to form this understanding, which may lead to limitations such as incomplete exploration and inadequate contextual support.

C2 Difficulty in understanding connections between isolated facts. While many current visualization methods are effective at uncovering more individual insights, they often lack mechanisms to establish connections between isolated findings.

3.2. Design requirements

To address the challenges identified in Section 3.1, we establish the following design requirements for FactExplorer based on a comprehensive survey of the literature on EDA.

First, FactExplorer should *automatically extract and organize data facts for a multifaceted overview* to address C1.

R1 Automated extraction of rich data facts from raw data. Data facts serve as the foundation for analysis. However, the process of manually extracting and organizing facts from raw data can be so tedious that it is almost impossible to achieve. Therefore, FactExplorer should support automated extraction of data facts from the input dataset (Ding et al., 2019; Lu et al., 2021; Wang et al., 2020), ensuring that the generated fact set covers an extensive range of data, and each data fact should include certain informational content (i.e., data items). Furthermore, it is essential to ensure that all facts correctly encode data attributes to visualization channels following empirical knowledge (Mackinlay et al., 2007; Wongsuphasawat et al., 2016, 2017).

R2 Hybrid representation of data facts to build multi-angle overviews. To provide meaningful overviews for guiding exploration, FactExplorer should extract and visualize salient semantic features from each data fact (Xie et al., 2021; Zhao et al., 2022). These features should capture the key characteristics and relationships between facts, enabling analysts to gain a comprehensive understanding of the fact space. Moreover, FactExplorer should allow users to interactively assign weights to these semantic features, giving them the flexibility to customize the overview based on their preferences and specific analytical goals (Xu et al., 2018).

Second, the system should *identify bridging connections between facts to facilitate the understanding of isolated facts* to tackle C2.

R3 Automatic generation of transition paths between facts. Understanding the connections between facts is crucial for analysts to form more synthesized and profound insights (Cao et al., 2020; Hullman et al., 2013; Kim et al., 2017). Therefore, FactExplorer should build transition paths to reveal how the analyst-specified facts are related and transformed, helping them establish a coherent understanding of the relationships between isolated findings.

Third, the system should also incorporate additional requirements focused on supporting the core aspects of *fact-based exploration* and *convenient user interaction*, which are also important for building a well-integrated EDA system.

R4 Topic-based drill-down to navigate fact-exploration. To assist analysts in exploring specific facts of interest within the overview, FactExplorer should automatically identify themes within the data facts and organize them accordingly (Li et al., 2020; Xu et al., 2018). This topic-based drill-down approach enables analysts to discover related facts in a structured and systematic manner, gradually deepening their understanding of the data (Joglekar et al., 2019).

R5 Convenient interactions to support flexible exploratory analysis. Flexible and convenient interactions are essential for supporting users in engaging with various analytical workflows (Xu et al., 2018; Zhao et al., 2022). To this end, FactExplorer should offer a range of interactive widgets and tools that enable users to dynamically manipulate and explore the data based on their specific questions and hypotheses. These features should include the ability to filter, inspect, and edit facts, with real-time updates to ensure that user adjustments are immediately reflected in the visualizations.

3.3. System framework

To achieve the design requirements discussed in Section 3.2, we present a fact-based analytical framework that employs an overview-to-detail analysis paradigm. As depicted in Figure 2, this framework is structured around three primary modules: fact extraction, fact embedding, and fact exploration.

3.3.1. Fact extraction

This module aims to identify hidden patterns within raw data and encapsulate them into discernible and actionable data facts (R1). By doing so, it prepares the data foundation for subsequent modules. While implementing this module is built upon well-established research (Ding et al., 2019; Lu et al., 2021; Wang et al., 2020), it also allows users to control the granularity of fact extraction (R5). Once set up, the module functions autonomously in the back-end without requiring complex user operations or additional inputs.

3.3.2. Fact embedding

In this module, we utilize a hybrid embedding approach to convert data facts into semantic vectors, which are then visualized on a 2D scatterplot. This hybrid representation incorporates logical and visual features, extracted through separate pipelines. The resulting vectors are combined using a weighted summation. This process allows users to prioritize one of the features they deem interesting, or relevant to

their analysis, thus highlighting its presence and importance in the overall visual representation (R2&R5).

3.3.3. Fact exploration

This module offers customized tools for conducting an in-depth exploration of data facts. While it integrates several flexible features, such as filtering and emphasizing specific facts according to various criteria, its most notable components are as follows. The first is a topic-based drill-down feature, which leverages a predefined hierarchical structure to enable fast and precise navigation to facts under a specific topic of interest (R4). The second function is the transition path generation, designed to enhance a user's comprehension of the connections between two specific facts by elucidating the evolutionary pathway from one fact to another (R3).

3.4. System interface and interaction

In this section, we introduce the user interface of FactExplorer and explain how users leverage its features for analysis. As illustrated in Figure 3, the interface consists of four primary components. The navigation view (Figure 3(a)) offers a quick overview of the statistical properties of the fact collection. It visualizes the distribution of facts across various task-specific categories and their visual encoding types. This view also allows users to adjust the weights of fact embeddings and modify specific fact elements. The fact overview view (Figure 3(b)) presents a scatterplot where semantic vectors of facts are projected. Each dot represents an individual fact, differentiated by color to indicate its type and scaled by size to reflect its significance score. The arrangement of facts within this view can be dynamically adjusted by changing the embedding weights, as demonstrated in sub-views Figure 3(b1–b3). The inspecting and editing view (Figure 3(c)) provides an interactive platform for users to explore specific facts in detail and record their observations. The transition path searching view (Figure 3(d)) displays the connections between selected facts, elucidating their relationships and interactions. Finally, the interface also includes a toolbar (Figure 3(e)) that gives users the flexibility to select different datasets and apply filters to facts based on their specific analytical needs.

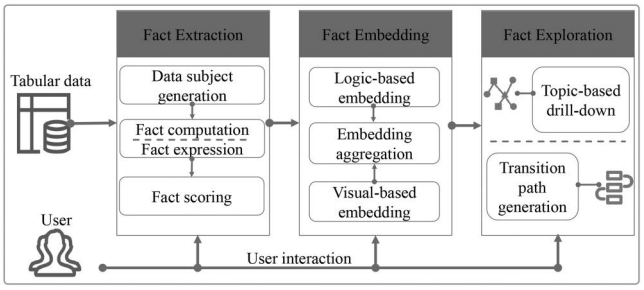


Figure 2. The framework of FactExplorer consists of three modules: Fact extraction, fact embedding, and fact exploration.

4. Fact extraction

Currently, we follow a general pipeline that has been widely used in previous works on data fact generation (Ding et al., 2019; Lu et al., 2021; Wang et al., 2020). The idea behind this extraction method is to first divide the data subjects (including data subsets and observed dimensions) from the raw data. Data facts are then computed within these data subjects, followed by their respective scoring and assembling them into visual representations. To better demonstrate the

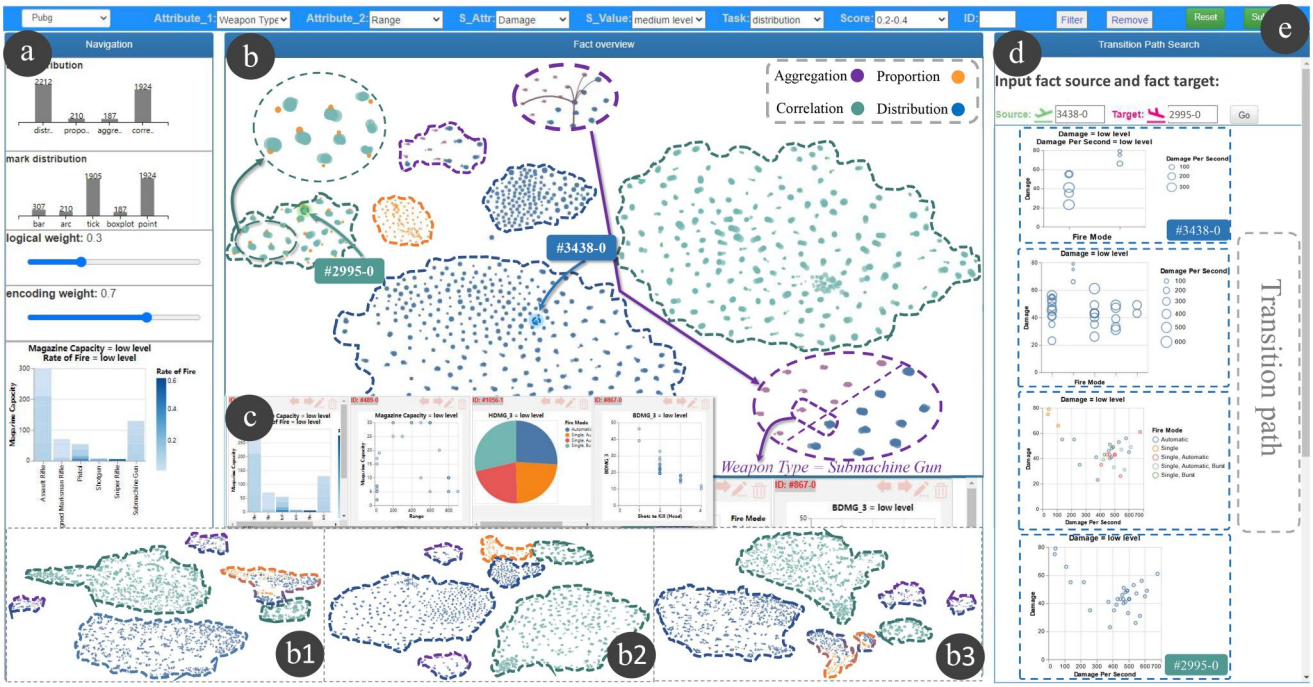


Figure 3. The interface of FactExplorer, which includes four major views: (a) navigation view, (b) Fact overview view, (c) Fact inspecting and editing view, and (d) Transition path searching view. (b1), (b2), and (b3) are additional overviews with varied embedding weights compared to (b). (e) is a toolbar that provides the features to select the analyzed dataset and filter facts.

Table 1. Table on vaccine and disease by country and year.

Country	Region	Year	DPT vaccine rates[a]	DALY[b]
Afghanistan	Asia	1990	0.25	104103
Albania	Europe	2004	0.97	27187
Angola	Africa	2002	0.47	82514

[a]The DPT vaccine is a combination of vaccines against three infectious diseases.
[b]DALY is a way of measuring disease burden.

process, a simplified table on *vaccine* and *disease burden*[2] (as shown in Table 1), which contains 5 attributes (*Country, Region, Year, DPT vaccine rates, and DALY*) and 4,312 records, is used as a running example.

4.1. Data subject generation

The fact extraction process begins with identifying the data subject. The data subset is obtained by progressively applying filters to split the data in a top-down structure. For categorical and temporal columns, filters specify particular values (e.g., "Year = 2016" in fact , of Figure 4(a)), while for numerical columns, filters correspond to bin ranges (e.g., "DALY = low level" in facts and of Figure 4(a)). Notably, over-filtering or applying too many bins can increase processing complexity and produce "invalid" data subsets containing only one or even no data items. The system counts the number of data items within each generated data subset, eliminating those with fewer than two, as the facts derived from them usually fail to reveal useful patterns for audiences. Furthermore, our preliminary experiments showed that filters have a more significant impact on the results than bins, especially for tables with more categorical and temporal attributes (see supplementary material for details on the results of fact extraction experiments). While adding more filters may generate more facts, it significantly reduces the extraction efficiency (which refers to the ratio of the number of valid data subsets to the total number of enumerated data subsets) and increases program execution time from tens of seconds to several minutes or even hours. In contrast, the impact of bins on processing time and efficiency is less pronounced. For larger datasets (rows > 1,000) with numerous numerical attributes, increasing bins can help uncover more fine-grained facts. However, excessive bins might be counterproductive for smaller datasets (rows < 100). Considering the trade-offs between the extraction depth, extraction efficiency, and computational efficiency, the system defaults to using two filters and three bins. Another reason for choosing three bins is that they also align with the general human perception of low, medium, and high levels. Overall, this practical compromise fits most situations but can be adjusted based on specific needs.

Once the data subset is established, various combinations of data attributes are identified as observation dimensions for analysis. After reviewing the literature on automatic visualization (Lu et al., 2021; Shi et al., 2021; Wang et al., 2020), we found that existing visualization design guidelines are typically applicable when there are no more than three dimensions. This is because the visualization techniques

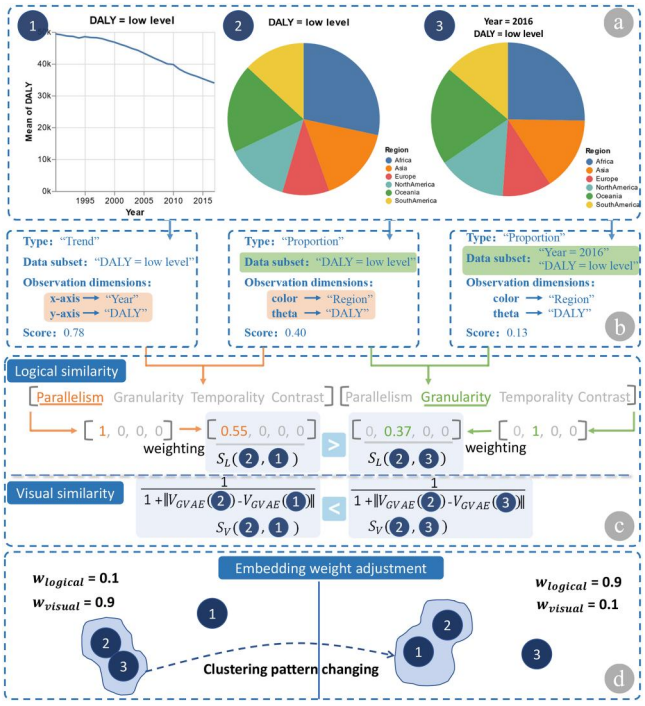


Figure 4. The examples of extracted facts and a demonstration of fact embedding. (a) shows the visual representation of three examples of extracted facts; (b) depicts the key components of the facts presented in (a); (c) demonstrates the computation of similarity between two facts; (d) illustrates how the distribution of the facts changes with different feature weights.

commonly used (e.g., bar chart, line chart, and scatterplot) are most effective for visualizing and analyzing three or fewer dimensions. Exceeding this limit may necessitate customized and complex visualization methods, which increase the difficulty in design and implementation and pose a higher cognitive burden on users. Thus, considering these factors, we limited the number of observation dimensions to three.

4.2. Fact computation

Once a data subject is given, we compute its applicable facts, including *Distribution*, *Correlation*, *Trend*, *Proportion*, and *Aggregation* (see supplementary material for details on these fact types), all of which are commonly used and proven utility in data analysis (Amar et al., 2005; Chen et al., 2009). The selection of fact types is guided by the combination of observation dimensions. For instance, *Trend* facts can be generated from a combination of numerical and temporal attributes, while *Correlation* facts are derived from a blend of two numerical attributes. These fact types are then mapped to appropriate visualization forms (e.g., a bar chart for *Distribution*, a line chart for *Trend*, and a scatterplot for *Correlation*). The mapping process also involves associating observation dimensions with specific visual channels (e.g., numerical dimensions typically find their place on channels like the y-axis, while categorical and temporal dimensions are presented on channels like the x-axis). These mapping guidelines have been pre-configured and are anchored in heuristics previously applied in systems like Show Me

(Mackinlay et al., 2007) and Vovager (Wongsuphasawat et al., 2016, 2017).

4.3. Fact scoring

This scoring is intended to help analysts prioritize their efforts on potentially more informative or interesting facts, thereby improving the efficiency of fact exploration. Two proven metrics are selected to score the interestingness of facts (Ding et al., 2019; Shi et al., 2021; Tang et al., 2017; Wang et al., 2020). One metric is significance, which measures the statistical characteristics of the fact, i.e., facts with more distinct statistical features (e.g., extreme deviations, significant growth or decline, and strong correlation) tend to receive higher significance scores. For instance, in Figure 4(a), fact has a higher score than due to its significant decrease, while has a relatively average distribution. Impact quantifies the data density of the fact, with facts containing more data items considered more important. As illustrated in Figure 4(a), facts and have similar proportional distributions, but has a higher score because it has more data items.

4.4. Fact expression

Finally, the computed facts are communicated to users through visual expressions and textual descriptions. Vega-Lite (Satyanarayan et al., 2017) is utilized here as the underlying visualization grammar for its concise declarative syntax, which offers a more streamlined way of creating detailed visualizations than programming methods like D3 (Bostock et al., 2011). Specifically, parameters (e.g., *mark*, *encoding*, and *transform*) within the Vega-Lite specifications are set based on the content of the data facts. For instance, *Trend* facts might be showcased using a line chart, denoted by (*mark*: "line"). Through *encoding* property, the observation dimensions of facts are mapped to the appropriate visual components (e.g., the x-axis, y-axis, color). Filters in the facts are reflected using the *transform* function to spotlight specific data points. Additionally, concise textual descriptions are added to enhance the understanding of the facts. These textual explanations are generated based on templates drawn from previous works (Jiang et al., 2021; Shi et al., 2021; Wang et al., 2020).

5. Fact embedding

To create a comprehensive fact overview, it is crucial to characterize facts using suitable models or representations. Recognizing the need to offer users a multifaceted perspective on facts, we employ a joint embedding approach that incorporates both logical and visual features. These features are selected based on their effectiveness in representing facts, as highlighted in prior research (Cao et al., 2022; Kim et al., 2017; Shi et al., 2021; Zhao et al., 2022). The logical features capture the data lineage links between facts, while the visual features depict the inherent information related to the visual representation of the facts themselves. Given the distinct characteristics of these features, separate embedding modules are designed for each, and their outputs are subsequently combined to form the final fact vector. Furthermore, users are empowered to interactively modify the feature weights, allowing them to adjust the balance between logical ties and visual representation in the embedding per their analytical needs.

5.1. Logic-based embedding

The logic-based embedding module (Figure 5-top) is engineered to encode logical relationships between facts at the data subject level, aiming to position facts that share logical connections closer in the embedding result. Drawing from relevant literature on inter-fact relations (Cao et al., 2020, 2022; Hullman et al., 2013; Shi et al., 2021), we identified four key relationships: *Parallelism*, *Granularity*, *Temporality*, and *Contrast*. Definitions and detection rules for these relationships are detailed in Figure 6. Compared to data-driven methods that rely on manually labeled data, our rule-based approach provides interpretability of the embedding results and is easy to implement and extend to incorporate more logical rules.

Specifically, the logic detection unit evaluates the logical similarity of fact pairs by detecting their logical connections based on predefined rules (Figure 6). The detection mechanism is detailed in Equation (1).

$$S_{logic_i}(a,b) = \begin{cases} 1, & if \quad logic_i \quad exists \\ 0, & if \quad no \quad logic_i \end{cases} \tag{1}$$

Herein, the logical similarity denoted as $S_{logic_i}(a,b)$, is a binary value that indicates the presence (1) or absence (0) of a specific logical relationship ($logic_i$) between two facts (a and b). The overall logical similarity $S_{logic}(a,b)$ between two facts is a collection of individual logical similarities, as indicated in Equation (2).

$$S_{logic}(a,b) = \left[S_{logic_p}(a,b),\ S_{logic_g}(a,b), S_{logic_t}(a,b),\ S_{logic_c}(a,b)\right] \tag{2}$$

For instance, as depicted in Figure 4(b,c), both facts and focus on the same data subset but select different observation dimensions for analysis. Thus, *Parallelism* is the only logic identified between them, and their logical similarity S_{logic} (,) is quantified as [1, 0, 0, 0].

To precisely measure the logical similarity, it is crucial to consider the coherence of each logic type (Cao et al., 2022; Shi et al., 2021). Equation (3) outlines the procedure for assigning weights to these logical relationships.

$$S_{weighted_logic}(a,b) = \left[w_p * S_{logic_p}(a,b),\ w_g * S_{logic_g}(a,b), w_t * S_{logic_t}(a,b),\ w_c * S_{logic_c}(a,b)\right] \tag{3}$$

In this formulation, $S_{weighted_logic}(a,b)$ denotes the weighted logical similarity and is derived by multiplying the original logical similarity with its corresponding weight (e.g., w_p, w_g). Figure 6 presents concrete values of these logic weights, which draw inspiration from Calliope's empirical study on narrative logic in data stories (Shi et al., 2021). Nevertheless, our approach introduces certain adjustments

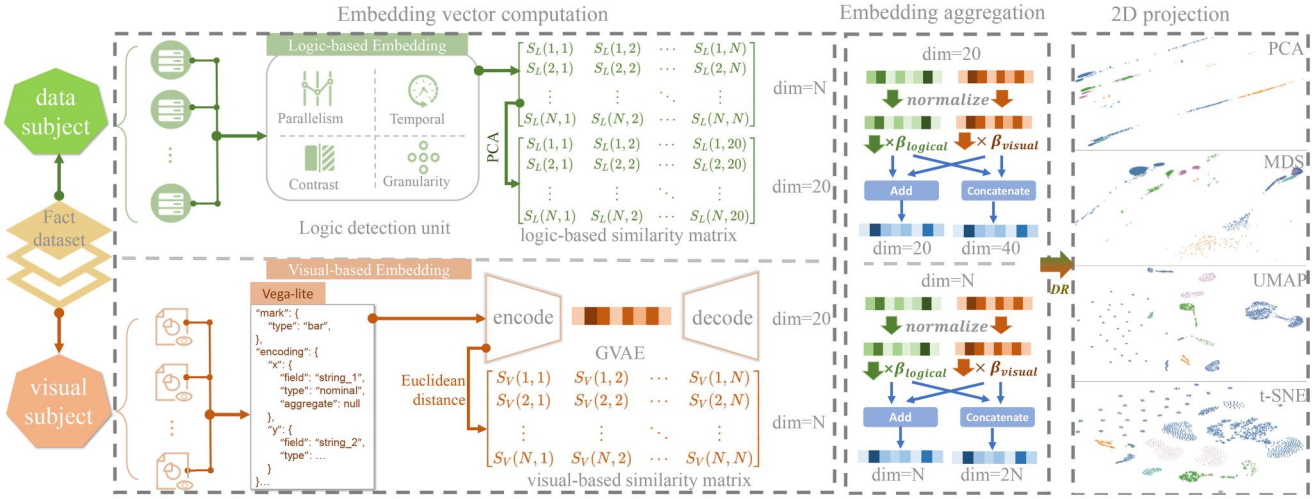


Figure 5. The demonstration of embedding models (including logic-based embedding, visual-based embedding, and embedding aggregation) and comparison of different DR methods.

to this foundation. Specifically, we combined the weights for "Elaboration" and "Generalization" into a unified *Granularity* category, as they both emphasize distinctions in data granularity. Furthermore, the "Cause-Effect" logic from Calliope was omitted due to its vague detection boundaries compared to other logical relationships. An example is displayed in Figure 4(c), where for fact , the *Parallelism* logic with fact exhibits stronger logical consistency compared to the *Granularity* logic with fact . In other words, when users observe fact , they prefer to explore facts linked through *Parallelism* logic (i.e., fact). Consequently, their weighted logical similarities are recorded as [0.55, 0, 0, 0] and [0, 0.37, 0, 0].

The final logical similarity between any two facts is denoted as S_L, which is computed by summing the values in $S_{weighted_logic}$ array. For instance, S_L (,) is calculated as 0.55, while S_L (,) is 0.37. Each fact undergoes the logic detection process against the others sequentially, producing an N-dimensional vector, where N is the total number of facts. Each dimension in the vector captures the logical similarity with a corresponding fact. This process constructs an $N \times N$ logic-based similarity matrix (Figure 5).

5.2. Visual-based embedding

In the visual-based embedding model (Figure 5-bottom), our system processes the visual representation information of facts provided in Vega-Lite format and converts them into corresponding visual vectors. Initially, the graph-based approach known as GraphScape (Kim et al., 2017) was considered. This method treats chart specifications as nodes within a graph, where edges represent the transformation costs between charts. The similarity between the two charts is calculated by summing the edge weights along the shortest path connecting them. However, these graph-based methods often depend on computationally intensive operations for similarity calculations and require perceptual experiments or user behavioral data to establish edge weights.

Beyond graph-based methods, machine learning-driven embedding techniques are another emerging approach. Inspired by the recent advancements in this field (Kusner et al., 2017; Zhao et al., 2022), we have chosen to implement the GVAE to embed visual features of facts due to its several advantages. First, GVAE employs an unsupervised learning mechanism with an encoder-decoder architecture, eliminating the need for predefined rules or manually labeled data. Second, GVAE is flexible and widely applicable. Once the model learns the data features, it can be easily incorporated into various applications. Finally, the availability of GVAE's open-source code facilitates our implementation.

The GVAE architecture consists of an encoder featuring a 3-layer Convolutional Neural Network (Gu et al., 2018), a densely connected layer with 256 neurons, and a decoder employing a 3-layer Recurrent Neural Network based on Gated Recurrent Units (Cho et al., 2014). Both the encoder and decoder have hidden dimensions of 256, and the latent vector space is characterized by 20 dimensions.

5.3. Embedding aggregation

To effectively integrate logical and visual features into a unified representation, we first need to resolve the dimensional discrepancies between the two types of feature vectors, which can otherwise lead to "unfair" weighting. This can be achieved through two alternative alignment approaches: (1) 20-dimensional, which employs PCA to reduce the dimensionality of the N × N logic-based similarity matrix to match that of the visual features (i.e., 20 dimensions); (2) N-dimensional, which computes an N × N visual-based similarity matrix based on the 20-dimensional visual features generated by GVAE. The visual similarity S_V between two facts a and b is calculated using the Euclidean distance between their GVAE-based features $V_{\mathrm{GVAE}}(a)$ and $V_{\mathrm{GVAE}}(b)$, as formulated in Equation (4).

$$S_V(a,b) = \frac{1}{1 + ||V_{\mathrm{GVAE}}(a) - V_{\mathrm{GVAE}}(b)||} \tag{4}$$

Logical relationship between facts	D	C	T	P	A
Parallelism ($logic_p$) reveals the logical parallelism between two facts that analyze the same data items from different perspectives, as identified by their shared data subset and varying observation dimensions or aggregation operations.	51.31%	26.18%	29.36%	54.96%	47.60%
Granularity ($logic_g$) refers to the degree of detail or specificity of the data collection analyzed within facts, and this relationship is established when one fact's data subset is a part of the other fact's data subset.	36.02%	34.88%	34.35%	37.34%	43.11%
Temporality ($logic_t$) signifies the chronological order of facts and is determined by the presence of varying temporal filters that distinguish the facts in terms of their position in time.	12.67%	26.18%	29.36%	7.70%	9.29%
Contrast ($logic_c$) emphasizes the comparison of two facts across different data items. The two facts with this relationship share the same fact type and observation dimensions but differ in the analyzed data subsets.	0.00%	12.75%	6.94%	0.00%	0.00%

Figure 6. The definition of four logical relationships and the likelihood distribution of these relationships occurring subsequently for different fact types-Distribution (D), Correlation (C), Trend (T), Proportion (P), and aggregation (A). The colors in the heatmap map the magnitude of the probabilities (greener indicates higher probability).

Upon aligning the features, we consider two fusion methods for combining them into a comprehensive representation: (1) addition-based as indicated in Equation (5), which employs element-wise summation to maintain the same dimensionality as the original features; (2) concatenation-based as illustrated in Equation (6), which joins the two features to generate a combined feature vector with twice the dimensionality of the individual features.

$$S(a,b) = w_{logical} \times S_L(a,b) + w_{visual} \times S_V(a,b) \quad (5)$$

$$S(a,b) = concat(w_{logical} \times S_L(a,b), w_{visual} \times S_V(a,b)) \quad (6)$$

In both equations, $S(a,b)$ represents the composite similarity between the facts a and b. The weights for the two features, $w_{logical}$ and w_{visual}, are set to 0.5 by default but can be modified interactively according to user preferences. Before fusion, both similarity features undergo normalization to ensure that neither feature disproportionately influences the combined outcome. Here, we employ the widely adopted 0–1 value normalization method.

Next, we conduct comparative experiments to analyze the differences among the four feature fusion methods obtained by combining the dimensionality alignment approaches with the fusion operations (Figure 5). Furthermore, we evaluate the performance of different DR methods in generating summarized visualizations. From these experiments, we derive the following key findings:

- Comparison of DR methods. Our preliminary experiments with various DR techniques showed that t-SNE and UMAP outperform MDS and PCA, especially in their ability to preserve local data structures (Figure 5). It is unsurprising because nonlinear DR methods generally excel in this area. Upon further in-depth comparisons, t-SNE performed better in terms of distance preservation and less visual cluttered than UMAP, aligning with the findings in (Espadoto et al., 2021). Moreover, although both methods achieved favorable outcomes through parameter tuning, t-SNE was less sensitive to hyperparameters in our scenarios.
- Characterization of feature types. We identified notable differences in the visualization outcomes between the two types of features. Features derived from visual content tend to be more scattered and show weaker clustering tendencies (e.g., Figure 7- and). In contrast, features grounded in logic exhibited clear and distinct clustering patterns (e.g., Figure and). This discrepancy may originate from the different embedding approaches used, with visual feature embeddings from the GVAE model introducing more uncertainty than those from rule-based logical embeddings.
- Impact of feature alignment. Our experiments also indicated no significant variations in the visual overviews produced using GVAE-based visual features and their associated similarity matrices (e.g., Figure 7- and). Conversely, the original logic-based similarity matrix demonstrated a stronger clustering structure than the PCA-processed one (e.g., Figure 7- and). This discrepancy may be due to information loss during DR, which weakens the structure among data points. Another finding also supports the viewpoint: the clustering effect of N-dimensional alignment methods is superior to that of 20-dimensional methods (e.g., Figure 7- and).
- Effects of feature fusion. While the concatenation-based fusion method preserved more original feature information, we observed no marked difference in visualization results between this method and the addition-based fusion method (Figure 7-top and bottom). This similarity suggests that increasing the feature dimensionality through concatenation does not necessarily enhance the discriminatory power of the resulting feature space, as there may be redundancy between the combined features.

Based on these findings, the system now defaults to using N-dimensional alignment and addition-based fusion operations for feature aggregation. It employs t-SNE as the default DR, with an optimized parameter set (perplexity of 25, early exaggeration of 12, and other parameters consistent with t-SNE's defaults). An example in Figure 4(d) demonstrates

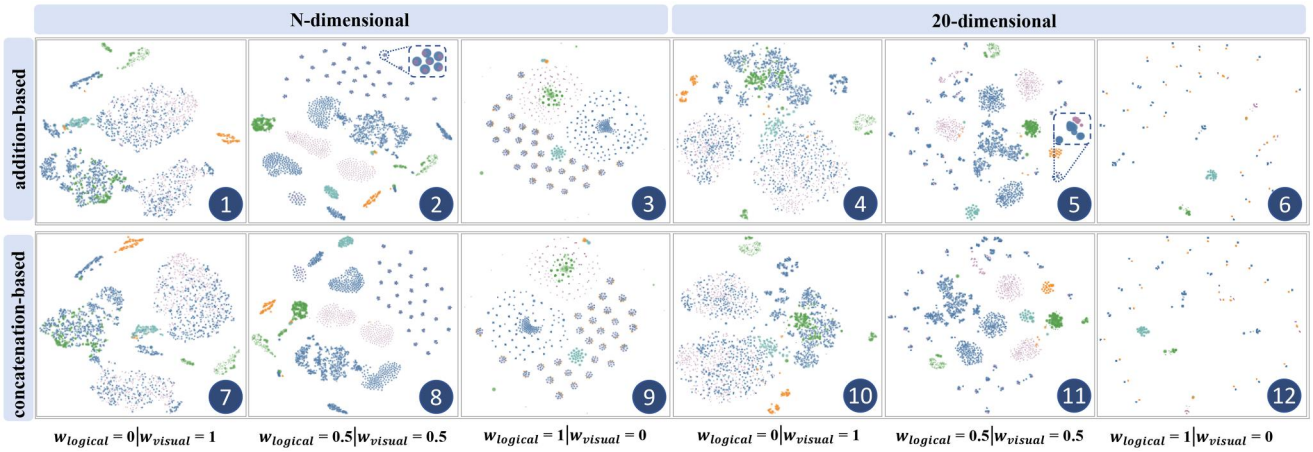


Figure 7. The comparative results of different feature fusion methods. The upper and lower parts present the outcomes of addition-based and concatenation-based fusion operations. The left and right sides detail the results of N-dimensional and 2-dimensional alignment methods under various weight ratios.

how varying the weight between visual and logical features impacts data clustering. With a higher weight on visual features, visually similar facts cluster together. Increasing the weight on logical features shifts the clustering towards facts with closer logical relationships.

6. Fact exploration

While overviews effectively capture the global characteristics of data facts, delving deeper into their relationships often requires enhanced interactive capabilities. To address this need, we introduce two tailored techniques: topic-based drill-down and inter-fact transition path generation.

6.1. Topic-based drill-down

Drill-down (Joglekar et al., 2019), a widely used technique in exploratory analysis, enables users to traverse predefined data hierarchies to uncover additional information. Building upon this concept, we present a topic-based drill-down method that provides a structured framework for navigating facts using pre-established thematic categories. Next, we elaborate on the definition of these topics and their role in supporting user exploration.

- *Task-based topic* Under this topic, facts share the same data items and observation dimensions but vary in task type. Diverse tasks convey distinct insights, and users gain a holistic understanding when they delve deeper into this topic.
- *Attribute-based topic* Within this category, facts have the same task type and data items while differing in observation dimensions. This category assists in scrutinizing various attributes of the same data items for a given task.
- *Subspace-based topic* Facts under this topic align in task type and observation dimensions but involve different data items. This topic allows users to analyze different sets of data items based on specific dimensions and tasks

When users right-click on a fact of interest, the system automatically retrieves associated facts based on these topics. Upon selecting a topic, the system labels eligible facts and visualizes their expansion pathways using curved lines. These labels and curves are commonly employed to represent the origin-destination path for intuitive visual highlights. To maintain focus and reduce visual clutter, a retracting mechanism is implemented, automatically collapsing the previous level of expansion as users drill down further.

6.2. Transition path generation

Transition path generation assists users in connecting seemingly unrelated facts by establishing well-defined paths between them. Initially, we approached this task as an optimization problem, aiming to maximize the logical coherence, expressive diversity, and significance of transition paths. However, this approach proved computationally expensive, with a time complexity of $O(n^n)$. While the applications of constraints and sampling showed promise in reducing the search space, they failed to deliver satisfactory results in practice. To balance efficacy and speed, we adopted a greedy algorithm with three stages:

1. Difference computation. We compute the differences between user-provided facts by comparing each keyword with its corresponding value in their Vega-Lite specifications. This comparison produces a set of operations that indicate how to transition from one fact to another. As illustrated in Figure 8 (top), an operation set contains different channels (e.g., mark, transform, and encoding) alongside their respective operations (e.g., modify, remove, add, and transpose).
2. Path generation. Building on the identified operations, we proceed to generate transition facts by executing each operation sequentially. This process continues to generate transition data facts until all operations are performed and the target fact is achieved. The primary challenge is to determine the most effective sequence of operations, which we address by prioritizing operations based on their associated costs. Operation costs are estimated using pairwise comparisons from prior

research, specifically leveraging cost rankings established by GraphScape (Kim et al., 2017). Their findings delineate the cost hierarchy as: *transpose>add*, *remove>modify* and *mark type<data transform<encoding operations*.

To further improve user recognition and memorization of fact transitions, a mark queue and a handler are employed to ensure diversity in the visual expression of transition paths. When a new transition fact is created, the handler checks if its mark information is already in the mark queue. If a duplicate mark is detected, the handler replaces the current mark with a randomly selected alternative. The modified information is then updated in the mark queue, maintaining synchronization between the queue and the generated transition facts. Figure 8 illustrates an example, showing the transition from fact to the target fact . Although facts , , and are of the same line type, they exhibit distinct visual expressions due to the mark adjustments applied during the path generation.

3. Fact rectification. The final step involves scrutinizing each fact in the transition path to ensure compliance with well-established guidelines. We employ a rule-based approach due to the advantages of its detection accuracy and reliability. The detection and rectification rules are built on insights from previous efforts (Chen et al., 2022; Moritz et al., 2018) and allow for user-customized adjustments. As an illustration in Figure 8, when a new temporal constraint ("Year = 2009") is added to fact , it conflicts with the "Year" indicated on the x-axis, the corresponding rectification method in this case is to remove the x-axis.

7. Usage scenario

To illustrate the effectiveness and usefulness of FactExplorer in aiding data understanding, we demonstrated a usage scenario featuring Tom, a 27-year-old digital media analyst with a Bachelor's degree and basic data analysis skills, familiar with tools (e.g., Tableau (2022), Power BI (2022)). In his daily work, Tom often needs to create digital posters with data from different areas (e.g., gaming and music) for promotional purposes. For this study, we selected a dataset[3] about firearms used in the popular shooting game, PUBG, which includes 37 firearm items, 3 categorical attributes (e.g., *Weapon Name*, *Weapon Type*) and 11 numerical attributes (e.g., *Damage*, *Magazine Capacity*, and *Range*). Tom was familiar with the gaming background of PUBG, such as how player success often depends on selecting appropriate weapons for diverse battleground scenarios (e.g., open-field environments like deserts or indoor environments like city streets). However, he was not familiar with the specifics of the data. Given this context, Tom's task was to freely explore the data using FactExplorer and attempt to identify interesting patterns that could help him create promotional posters in the future. Prior to the analysis, Tom received a brief 15-minute introduction to FactExplorer. Throughout his exploration, he voiced his thoughts following the think-aloud protocol, and guidance was provided when necessary. Tom's findings and comments were recorded to evaluate the efficacy of FactExplorer.

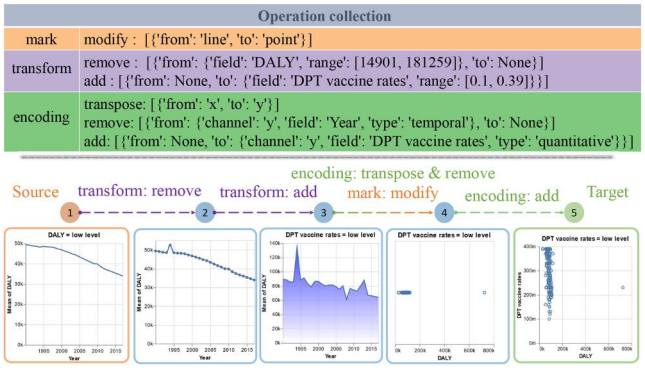


Figure 8. An example of transition path searching. The top is an operation collection and the bottom is the path generation process.

7.1. Inspect the overview of all facts

Tom began by inspecting the statistical overview of all facts generated by the system (Figure 3(a)), where Tom noted that *"it does not contain Trend facts due to the lack of temporal attributes in the dataset"* (R1). By adjusting the embedding weights, Tom observed how the relative positioning of facts was affected, as demonstrated in Figure 3(b1–b3). He stated, *"Upon adjusting the weights, certain previously scattered facts now cluster together"* (R2&R5). Focusing on a scenario with a logical weight of 0.7 and a visual weight of 0.3, Tom identified compelling patterns within the data fact landscape. For instance, multiple clusters of *Correlation* and *Proportion* facts about the same data subsets appeared on the left side of the overview (Figure 3(b)), as well as symmetrically arranged clusters of *Aggregation* and *Distribution* facts linked by the same filter conditions emerged at the top of the overview (Figure 3(b)).

In the above process, we observed that Tom could easily and proactively examine adjacent data facts within clusters. A simple click on the associated data points allowed him to access detailed information without the necessity of additional manual operations. Through these examinations, Tom rapidly developed a structured understanding of the data, e.g., identifying correlations among varying attributes within subsets. This capability benefits from the semantic representation of data facts, which allows facts with relevant information to cluster together, thereby forming localized structured knowledge.

7.2. Understand the association between isolated facts

Next, Tom delved deeper into the facts of interest to understand their associations. During this process, he encountered two seemingly contradictory facts. The first fact #3438-0, filtered by *Damage = low level* and *Damage Per Second = low level*, suggested a negative correlation between *Damage* and *Damage Per Second* (Figure 3(b)). In

contrast, another fact #2995-0, which was filtered only by *Damage = low level*, did not exhibit this correlation (Figure 3(b)). To resolve this inconsistency, Tom queried for a transition path linking the two facts (R3). From the results presented in (Figure 3(d)), he observed that omitting the filter *Damage Per Second = low level* introduced additional firearm items such as *Single Burst*, *Single Automatic*, and *Single Automatic Burst* under the category of *Fire Mode*. These additional firearm items disrupted the previously observed negative correlation between *Damage* and *Damage Per Second*.

7.3. Focus on a specified analytical task

Now, Tom focused on a specific question about identifying the most suitable firearm for open-field battlegrounds. In such settings, the primary consideration is a firearm's *Range*. By applying a filter that emphasized high *Range* values, he discovered fact #352-1 (Figure 9). From which Tom found that "*Only Design Marksman Rifle and Sniper Rifle qualified as long-range weapons, with the latter having superior range*" (R5). This finding was further validated by fact #352-0 (Figure 9), detailing that the *Sniper Rifle* exceeded the range of the *Design Marksman Rifle* by approximately 100 meters. Through an attribute-based drill-down, Tom found fact #14-0 (Figure 9), showcasing that the *AWM* stood out for its exceptional range within its category (R4).

To ensure a comprehensive assessment, Tom also identified *Shoots to Kill (Head)* and *Shoots to Kill (Chest)* as other critical metrics for assessing firearm effectiveness in open-field scenarios. By applying fact-filtering and attribute-based drill-down, he located facts (#557-0 and #289-0 in Figure 9) that provided insights into the performance of various weapon types and specific weapons in these metrics. Additionally, Tom explored other influential indicators, such as the distribution of *HDMG_3* (fact #34-0 in Figure 9), to further validate his findings. Finally, Tom summarized his findings and concluded that the AWM, with its exceptional range and commendable effectiveness in various metrics, stands out as the ideal firearm for open-field battlegrounds.

7.4. Summary feedback

After completing the exploration, we conducted a brief interview with Tom to gain further insights into his experience. He first noted that, from the perspective of facilitating EDA, he believed FactExplorer has more potential than traditional analytics tools, especially for professional data analysts. He explained, "*When using tools such as Tableau and Excel, my focus is on finding interesting dimensions, selecting appropriate visualizations to observe them, and iterating repeatedly. FactExplorer automates these basic data visualization operations, allowing me to devote more energy to advanced fact-based explorations.*" Moreover, another advantage Tom appreciated about FactExplorer is its fact overview. He stated, "*The fact overview provided by FactExplorer is a feature not available in previous tools. It comprehensively presents the underlying patterns in the data, offering richer context and guidance in completing my exploration.*" However, Tom also mentioned that the shortcoming of FactExplorer is that it demands more user knowledge and requires more time to learn and become familiar with, which may require further design optimizations to improve. Overall, Tom concluded that FactExplorer, with its powerful features and interactivity, has the unique value and potential to assist users in discovering insights more effectively and comprehensively.

8. User study

To further understand how FactExplorer enhances data exploration and facilitates insight discovery, we evaluated the FactExplorer system using quantitative experiments and a controlled user study. In particular, we first evaluated the effectiveness of our proposed key techniques (including fact embedding and transition path generation). Next, we conducted comparative experiments to analyze the overall performance of FactExplorer in enhancing data exploration and facilitating insight discovery.

8.1. Evaluation of the key techniques

We conducted an online test[4] to validate the consistency between fact-embedding results and user perceptions, as well as the effectiveness of transition paths in helping users understand isolated facts. The results demonstrate that our embedding technique can successfully capture the logical and visual semantics of facts, as well as the ability of our

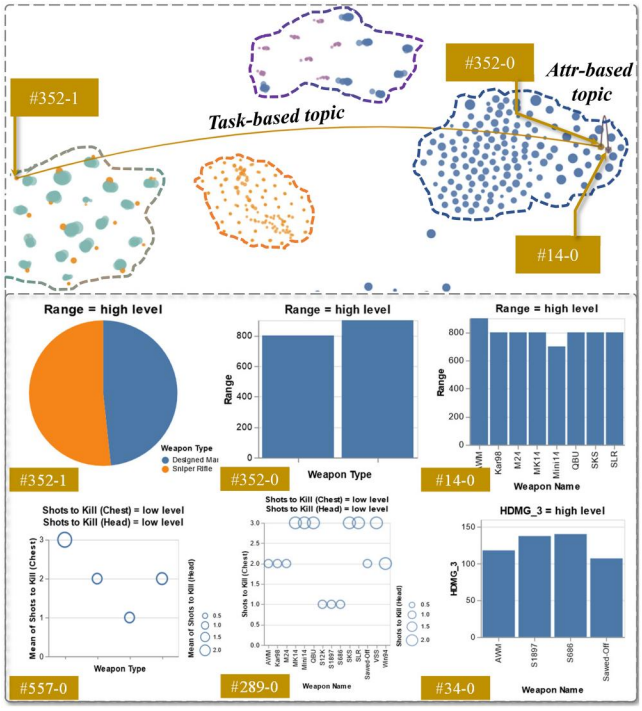


Figure 9. The upper part illustrates the process of topic-based drill-down exploration, while the lower part provides detailed information on specific facts.

transition path generation technique to elucidate the underlying connections between facts.

8.1.1. Design and procedure

We recruited 30 participants (11 female and 19 male) aged between 22 and 30 years (μ = 24.3 years, σ = 2.1 years) for this evaluation. Their backgrounds encompassed a wide range of expertise, e.g., data visualization, digital media, and computer vision. They had varied experience in data analysis, and 16 of them had more than one year. To ensure a comprehensive and fair assessment, we prepared three distinct datasets from different domains for testing, each with varying sizes and dimensions: *vaccine* and *disease burden* with 4,312 rows and 5 columns (introduced in Section 4), *PUBG's firearm* with 37 rows and 14 columns (introduced in Section 7), *women entrepreneurship and labor force*[5] with 51 rows and 8 columns. In addition, we also prepared a *Cars*[6] dataset with 303 rows and 9 columns for the participants' training. This test took approximately 25 minutes to complete, and participants received a compensation of about 4 USD for their time.

The evaluation began with a 15-minute introduction to relevant technical concepts and specific evaluation tasks. Next, we manually generated test scenarios for fact embedding and transition paths and required participants to complete the evaluation task. More specifically, we extracted five distinct embedding result segments for fact embedding, each accompanied by a projection scatterplot. Within each scatterplot, participants received detailed information on specific facts and were asked to identify the most plausible unknown fact from four candidates. Both accuracy and decision time were recorded as evaluation metrics. Similarly, we evaluated the transition path by presenting participants with five sets of selected fact pairs and their corresponding paths. Participants rated the quality of these paths, considering aspects such as *helpfulness in understanding specific facts*, *logicality*, and *visual expressiveness*. A 5-point Likert scale was used to collect their overall ratings. To further ensure a fair evaluation, we counterbalanced the fact types when constructing the test scenarios. Furthermore, we confirmed that no participant had prior exposure to any of the facts included in the evaluation tasks and encouraged them to fully understand the information before making decisions.

8.1.2. Results

Figure 10-top illustrates the differences in the completion times among various test groups, ranging from the fastest group T1 ($\mu = 21.67_{[19.1,\ 24.24]}$ seconds) to the slowest group T5 ($\mu = 39.55_{[35.93,\ 43.18]}$ seconds). A Kruskal-Wallis H-test suggested significant differences in completion times among the groups ($H = 55.04$, $p < .001$), which aligns with the varying complexity of the test scenarios. Despite the differences in completion times, all groups maintained high accuracy rates above 83.3%. A Chi-Square Test of Independence yielded a non-significant result ($\chi^2 = 5.2$, $p = .267$), indicating that scenario complexity had minimal impact on the precision of participants' interpretations. These results suggested that in most testing cases, the participants agreed that the logical and visual similarities between facts calculated by the system fit their perceptions.

The evaluation of transition paths also achieved positive results (Figure 10-bottom), each transition path received an average rating above 3.87. A Kruskal-Wallis H-test revealed no statistically significant differences in user scores across different transition path types ($H = 4.186$, $p = .381$). This result suggested that participants recognized the constructive nature and practicality of the transition paths.

8.2. Evaluation of FactExplorer's overall performance

To verify the capabilities of FactExplorer in enhancing analysis and facilitating insight discovery, we also conducted a comparative study with Foresight (Demiralp et al., 2017). The selection of Foresight was based on the following considerations. First, it is a well-established system in the visualization field, making it a reliable baseline. Second, both tools aim to support insight-driven data exploration, however, they employ different approaches (e.g., FactExplorer uses *embedding + scatterplot* for presenting and analyzing insights, whereas Foresight employs a *categorization and ranking* approach). This contrast allows for a meaningful evaluation of FactExplorer's capabilities. Unfortunately, we have not found its open source. As an alternative, we developed a simplified baseline system (Figure 11). To ensure fairness, we strictly replicated key features of Foresight in our baseline system implementation, such as the *classify + rank* exploration paradigm, heatmaps for characterizing attribute correlations, and other interactive features. Furthermore, the data facts and scoring metrics used in the baseline system are derived from FactExplorer. We established the following hypotheses:

H1 FactExplorer outperforms the baseline in facilitating knowledge discovery.

H2 FactExplorer receives higher user satisfaction than the baseline.

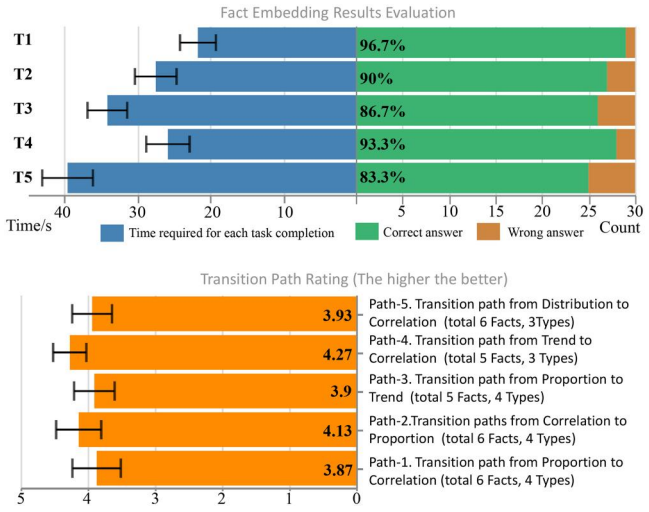


Figure 10. The evaluation results on fact embedding (top) and transition pathway ratings (bottom). Error bars represent 95% CI.

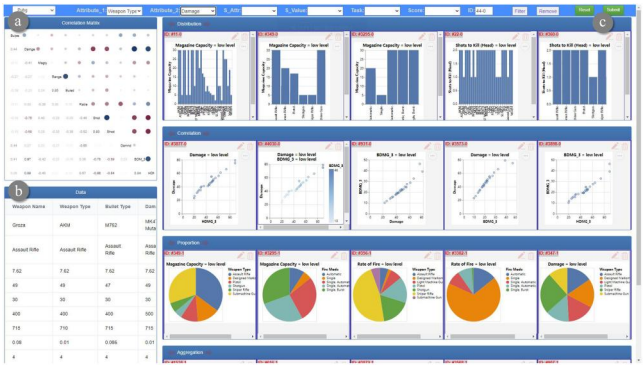


Figure 11. The user interface of baseline system. (a) Correlation matrix view, (b) data table view, (c) fact exploration view with categorization + ranking.

8.2.1. Design and procedure

We invited 16 participants (6 females and 10 males, age: μ = 25.6 years, σ = 2 years) from the previous evaluation round to continue participating in this assessment. These participants were selected because they had more extensive data analysis experience, which was necessary for this evaluation as it required in-depth usage of the system to complete exploratory analysis tasks. They were familiar with commonly used data analysis tools (e.g., ggplot (Valero-Mora, 2010), Maplotlib (2022)), and all had more than one year of experience in data analysis (μ = 3.2 years, σ = 1.3 years). For clarity, we numbered these participants from P1 to P16. In this round of assessment, we employed a within-subjects design. The entire assessment process lasted approximately 90 minutes, and participants involved in this round of evaluation received an additional compensation of about 12 USD.

First, we spent 20 minutes training the participants, including introducing the study design, providing detailed demonstrations of FactExplorer and the baseline system, and explaining user-related matters. Next, we asked the participants to explore *PUBG's firearm* dataset (introduced in Section 8.1) using both FactExplorer and the baseline system. To mitigate potential learning effects, we designed the experiment with randomized task sequences and sufficient time intervals between the use of each system. Throughout the process, participants were encouraged to think aloud and share their insights on the effectiveness, usability, and potential limitations of the systems. After the in-depth engagement and exploration, the participants completed a questionnaire using a 7-point Likert scale to rate various aspects and overall performance of the two systems.

Following this, the participants were engaged in evaluating the fact extraction results. Their review revolved around the following criteria: Task Type (*whether the task types of the facts meet their analysis requirements*), Visual Representation (*whether the visual presentation style of the facts is intuitive*), Fact Scoring (*whether the scoring of the facts aligns with their perception*), and Coverage (*whether the extracted facts sufficiently cover most of the important aspects of the data*). We collected participants' ratings on these metrics using a 5-point Likert scale.

8.2.2. Results

To evaluate *H1*, we tested the following null hypothesis *H0*: there is no difference in the exploration coverage between users of FactExplorer and the baseline system. Here, the exploration coverage (Sarvghad et al., 2017; Zeng et al., 2022; Zhao et al., 2022) was measured using three metrics: *data fact coverage*, *data variable coverage*, and *data subset coverage*. These metrics quantified the number of data facts, variables, and subsets that participants were interested in and focused on examining during their exploration. Higher coverage generally indicates more comprehensive exploration and more insights discovered. Correspondingly, we collected these exploration messages from the participants (Figure 12-left). The Mann-Whitney U tests revealed that FactExplorer significantly outperformed the baseline ($p < .01$) on all coverage metrics: *data fact coverage* ($\mu = 32.81_{[25.89,\ 39.73]}$ vs. $\mu = 13.94_{[12.33,\ 15.55]}$), *data variable coverage* ($\mu = 11.56_{[10.89,\ 12.24]}$ vs. $\mu = 9.5_{[8.55,\ 10.45]}$), and *data subset coverage* ($\mu = 15.19_{[13.13,\ 17.25]}$ vs. $\mu = 9.13_{[7.91,\ 10.34]}$). This superiority of FactExplorer is likely because its overview of data facts and fact-based analytical features efficiently help users identify contextually relevant facts, thereby motivating them to examine more data facts. Based on these findings, we rejected the null hypothesis *H0*, supporting our hypothesis *H1*.

To evaluate *H2*, we tested the following null hypothesis *H0*: there is no difference in user ratings between FactExplorer and the baseline system for any metrics (the scoring results are shown in Figure 12-middle). The Mann-Whitney U tests showed that FactExplorer received significantly higher ratings ($p < .01$) for several metrics, including *Functions provided by system* ($\mu = 5.31_{[4.74,\ 5.89]}$), *Overview perception* ($\mu = 5.94_{[5.44,\ 6.43]}$), and *Contextual EDA* ($\mu = 5.31_{[4.77,\ 5.85]}$) compared to the baseline ($\mu = 4.06_{[3.46,\ 4.66]}$, $\mu = 4.25_{[3.72,\ 4.78]}$, and $\mu = 4.25_{[3.68,\ 4.82]}$, respectively). However, for the metrics *Easy to learn* (FactExplorer: $\mu = 4.5_{[4.02,\ 4.98]}$ vs. baseline: $\mu = 5.06_{[4.57,\ 5.56]}$, $p = .13$) and *Easy to use* (FactExplorer: $\mu = 4.38_{[3.9,\ 4.85]}$ vs. baseline: $\mu = 4.88_{[4.36,\ 5.39]}$, $p = .49$), the differences were not statistically significant, and FactExplorer was rated lower than the baseline. This finding is not surprising, given that FactExplorer is a more complex system with richer interactive features, which may require more time for users to explore and familiarize themselves with. Overall, while the null hypothesis *H0* was rejected for certain metrics, supporting our hypothesis *H2*, it was not supported across all metrics.

We also analyzed participant ratings for additional features of FactExplorer, including *Drill-down helpful* ($\mu = 5.56_{[5.09,\ 6.04]}$), *Interaction fluency* ($\mu = 5.94_{[5.48,\ 6.39]}$), and *Visualization Intuitive* ($\mu = 5.44_{[4.85,\ 6.02]}$). Overall, participants provided positive feedback for these aspects of the system. These ratings suggested the effectiveness of the system in supporting users' analytical needs and providing a smooth and intuitive user experience. Moreover, participants' average ratings across various metrics for fact extraction consistently exceeded 3.8 out of 5 (Figure 12-right), including *Task type* ($\mu = 3.94_{[3.58,\ 4.3]}$), *Visual expression* ($\mu = 4.06_{[3.7,\ 4.42]}$), *Fact scoring* ($\mu = 3.88_{[3.55,\ 4.2]}$), and *Coverage* ($\mu = 4.31_{[3.94,\ 4.69]}$). These results suggested that

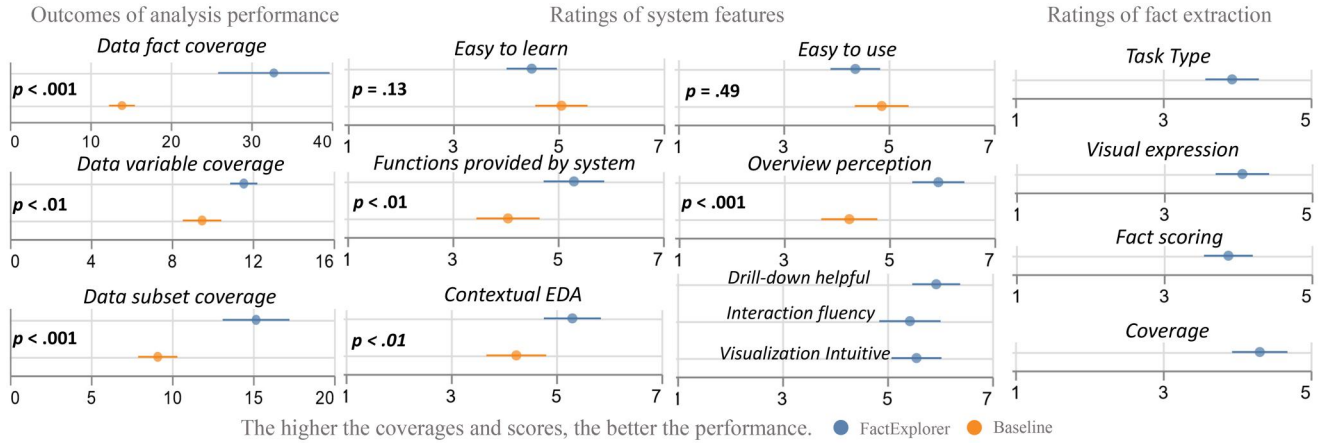


Figure 12. The results of participant analysis performance (left), system features ratings (middle), and fact extraction evaluation (right). Error bars represent 95% CI.

participants generally recognized the quality and effectiveness of the fact extraction techniques employed in FactExplorer.

8.2.3. Feedback

Following each session, participants were involved in concise interviews to gather their feedback and document their experiences with the systems. The interviews were conducted in Chinese, the native language of the participants, and the feedback collected was then translated into English for reporting.

8.2.3.1. Learning curve. First, participants' data analysis experience directly affected their ability to learn and utilize FactExplorer. For those who had extensive experience with visualization, they thought learning was not difficult, *"Although FactExplorer's interface appears complex, all the visual designs are familiar and intuitive"* (P1). In contrast, participants with limited experience in data analysis encountered a steeper learning curve when using FactExplorer. P15 commented, *"FactExplorer is a complex system with multiple views, which made it somewhat challenging for me to navigate."* This aligns with the collected ratings, as FactExplorer integrates more exploration support features, which to some extent increases the challenge of learning and usage.

8.2.3.2. System utilization. First, since the facts used in Foresight are directly sourced from FactExplorer, we only gathered user feedback on our existing fact extraction method. The majority of participants expressed appreciation for the utility of the automatic fact extraction feature and were satisfied with its results. Throughout the user study, no reports were received regarding problematic facts (e.g., incorrect visual coding, blank charts, etc.). Nonetheless, some participants (P2 and P5) highlighted the simplicity of the novelty score metric, suggesting that it could be enhanced by integrating more diverse evaluation metrics. Additionally, P4 pointed out the scalability issue, emphasizing the impact of data scale on computational load, and recommended *"implementing an incremental approach where the system first processes the most critical and relevant data fragments."*

As participants delved deeper into using the system for data exploration, they consistently acknowledged that FactExplorer provides more powerful analytical capabilities and contextual support compared to Foresight, which is reflected in the following aspects of feedback. In terms of providing a data overview, most participants found FactExplorer's fact overview scatterplot to be significantly more effective than Foresight's attribute correlation matrix. P1 stated, *"In the overview, I can immediately inspect and analyze interesting facts in place, whereas in the attribute correlation matrix, clues require additional analytical operations and view transitions for observation."* The same applies to the association analysis and understanding of facts. P7 remarked, *"In FactExplorer, the positions of facts in the scatter plot can be readjusted based on user requirements, while in Foresight, facts are organized and presented according to fixed categories."* P9 mentioned, *"The topic-based drilling function in FactExplorer can quickly associate related facts, but in Foresight, users need to perform complex operations and transitions to achieve this goal."* P11 added, *"The transformation path function helps me explain the connections between some facts, which is missing in the baseline."*

Although the participants found FactExplorer to offer richer and more powerful functionalities, they also proposed some suggestions for design and functional improvements. For instance, P4 pointed out, *"Currently, there is only one main overview view, which may not be conducive to comparing observations over time."* P1 suggested, *"Actively presenting thematic connections between key facts at the beginning of exploration can promote users' grasp of the fact overview."* P3, P7, and P8 recommended introducing an insight-sharing panel to facilitate the exchange of exploration discoveries.

9. Discussion

Here we discussed the lessons as well as the limitations that emerged from this work.

9.1. Lessons learned

Balancing guidance and independent exploration. Striking this balance is a significant challenge in designing data

analysis systems (Li et al., 2021; Vartak et al., 2017), but in this work, we found it is crucial to consider the audience's background and expertise in the process. From the user study, we observed that participants' reactions to FactExplorer varied significantly depending on their level of data analysis experience. Experienced participants swiftly engaged in the analysis process. Conversely, participants lacking experience sometimes felt overwhelmed and unsure how to proceed. Consequently, for novice users with limited data analysis experience, providing more structured guidance can be beneficial. On the other hand, experienced users can be more available for a more open-ended and flexible approach. Moreover, we discovered that over half of the participants sought clarification on how the facts were automatically produced. Thus, the explainability and transparency of the recommendation guidance are also important and need to be considered (Sperrle et al., 2023; Zhang et al., 2024).

Synthesized representation of visualization charts. Visualizations can take diverse forms such as text, images, and structured graphical representations (e.g., Vega-Lite) (Wu et al., 2022). In this work, we observed that selecting different perspectives for visualization embeddings can lead to features with distinct characteristics. However, most existing studies (Xu et al., 2018; Zhao et al., 2022), including ours, focus on limited perspectives and lack a systematic, comprehensive analysis to compare the similarities and differences between representations from various viewpoints. This is especially important when considering specific task scenarios (chart retrieval (Li et al., 2022; Nowak et al., 2024)), where different perspectives may have their own pros and cons.

9.2. Limitations and future work

9.2.1. Supporting more powerful fact extraction

Similar to prior work (Lu et al., 2021; Wang et al., 2020), FactExplorer utilizes a heuristic algorithm for fact extraction. While this approach provides controllable and interpretable outcomes, it has notable limitations. Primarily, it disregards the semantic importance of data fields, potentially producing statistically significant yet practically irrelevant findings. Enhancing the extraction process by integrating NLP techniques to better understand semantic relationships among data fields is a promising improvement. Scalability also poses a challenge, as the system struggles with complex data structures (e.g., hierarchical tables) and faces increased computational demands with growing data volume and dimensions. Effective strategies to address these issues include standardizing complex data into uniform formats, as proposed by P2, and implementing data sampling and prioritization techniques recommended by P3. Additionally, while the system measures fact interestingness, it lacks comprehensive metrics for overall extraction quality, which may undermine user trust and understanding. Therefore, further research and the development of holistic evaluation standards are necessary.

9.2.2. Creating rich and colorful visual expression

Developing diverse and engaging fact representations within a complex visual design space is inherently challenging. Currently, the system uses a template-based approach for visualization, which is efficient but limits presentation diversity. P2 noted, *"A broader range of visualization styles would be refreshing, rather than repetitive visual contexts for distinct datasets."* Additionally, the system's overview lacks visual guiding cues. P8 suggested, *"Integrating rose charts to highlight fact cluster features could provide deeper insights into fact distribution."* The overview also suffers from overlapping data points, hindering users' ability to explore finer details. Future versions will address these overlaps algorithmically while preserving the relationships among facts. Lastly, transitions between facts can be improved, as P11 stated, *"Adding animated transitions during fact switches could enhance the overall visual experience."*

9.2.3. Enabling more engaging fact exploration

A key feature of FactExplorer is the ability to adjust embedding weights, allowing diverse insights into fact distribution. However, uncertainty in weight adjustments can hinder precise pattern recognition. To address this, P5 suggested, *"Automatically identifying and recommending ideal weight combinations would streamline the discovery of underlying patterns."* Furthermore, P1 highlighted the need for guidance on drilling depth in the topic-based exploration process to prevent unproductive searches. Additionally, the system only supports transition queries between two specific facts. Future developments should expand its capabilities to derive insights from transition networks involving multiple facts. Finally, the system employs a traditional interaction paradigm. Further enhancements could include integrating additional modal interaction tools to enrich the user exploration experience (Amazu et al., 2024; Lingam et al., 2024; Zhang et al., 2024).

10. Conclusion

In this work, we introduced FactExplorer, a novel system that reshapes the traditional EDA process by shifting the investigation focus from raw data to data facts. The system automatically extracts potential data facts from the user-uploaded dataset and embeds them into a semantic vector space using a logic-visual hybrid representation for a fact overview. FactExplorer also offers powerful interactive exploration techniques to facilitate further exploration of data facts, including topic-based drill-down and transition path generation. The former allows users to query related facts through structured topics, while the latter helps users understand the connections between specified facts. Through a detailed usage scenario and a controlled user study with real-world datasets, we demonstrated the practical applications and effectiveness of FactExplorer. Our findings indicated that FactExplorer effectively facilitates comprehensive data exploration and helps develop synthesized insights. Finally, we discussed the

system's limitations and explored potential avenues for future development.

Notes

1. Open source: https://github.com/jiangqicd/factspace.
2. https://www.kaggle.com/datasets/arnavsharmaas/vaccine-and-disease-correlation-and-rates.
3. https://www.kaggle.com/datasets/aadhavvignesh/pubg-weapon-stats.
4. jiangqicd.github.io/survey/.
5. https://www.kaggle.com/datasets/babyoda/women-entrepreneurship-and-labor-force.
6. https://github.com/jiangqicd/factspace/blob/main/data/cars-w-year.csv.

Acknowledgments

We are grateful to all other participants for their participation in the discussions, suggestions, and evaluations. We also thank the reviewers for their insightful comments.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The work is supported by National Natural Science Foundation of China (62422607, 62372411, 62036009, 62432014), Zhejiang Provincial Natural Science Foundation of China (LR23F020003 and LTGG23F020005), and Fundamental Research Funds for the Provincial Universities of Zhejiang (RF-B2023006).

ORCID

Qi Jiang http://orcid.org/0000-0002-2732-1571
Guodao Sun http://orcid.org/0000-0002-8383-8153
Yue Dong http://orcid.org/0000-0001-9793-3744
Lvhan Pan http://orcid.org/0000-0002-8272-6096
Baofeng Chang http://orcid.org/0000-0002-1028-716X
Li Jiang http://orcid.org/0009-0000-2446-4000
Haoran Liang http://orcid.org/0000-0002-5906-1380
Ronghua Liang http://orcid.org/0000-0003-2077-9608

References

Abdi, H., & Williams, L. J. (2010). Principal component analysis. *WIREs Computational Statistics*, *2*(4), 433–459. https://doi.org/10.1002/wics.101

Amar, R., Eagan, J., & Stasko, J. (2005). Low-level components of analytic activity in information visualization. *Proceedings of IEEE Symposium on Information Visualization* (pp. 111–117). https://doi.org/10.1109/infvis.2005.1532136

Amazu, C. W., Mietkiewicz, J., Abbas, A. N., Briwa, H., Alonso-Perez, A., Baldissone, G., Fissore, D., Demichela, M., & Leva, M. C. (2024). Exploring the influence of human system interfaces: Introducing support tools and an experimental study. *International Journal of Human–Computer Interaction*, 1–18. https://doi.org/10.1080/10447318.2024.2376354

Borg, I., & Groenen, P. J. (2005). *Modern multidimensional scaling: Theory and applications*. Springer Science & Business Media. https://doi.org/10.1007/0-387-28981-x

Bostock, M., Ogievetsky, V., & Heer, J. (2011). D3: Data-driven documents. *IEEE Transactions on Visualization and Computer Graphics*, *17*(12), 2301–2309. https://doi.org/10.1109/TVCG.2011.185

Cai, H., Zheng, V. W., & Chang, K. C.-C. (2018). A comprehensive survey of graph embedding: Problems, techniques, and applications. *IEEE Transactions on Knowledge and Data Engineering*, *30*(9), 1616–1637. https://doi.org/10.1109/TKDE.2018.2807452

Cao, Y.-R., Li, X.-H., Pan, J.-Y., & Lin, W.-C. (2022). *VisGuide: Useroriented recommendations for data event extraction* [Paper presentation]. CHI Conference on Human Factors in Computing Systems (pp. 1–13). https://doi.org/10.1145/3491102.3517648

Cao, Y.-R., Pan, J.-Y., & Lin, W.-C. (2020). *User-oriented generation of contextual visualization sequences* [Paper presentation]. Extended Abstracts of ACM Conference on Human Factors in Computing Systems (pp. 1–8). https://doi.org/10.1145/3334480.3383057

Cavallo, M., & Demiralp, C. (2018). Clustrophile 2: Guided visual clustering analysis. *IEEE Transactions on Visualization and Computer Graphics*, *25*(1), 267–276. https://doi.org/10.1109/tvcg.2018.2864477

Ceneda, D., Gschwandtner, T., May, T., Miksch, S., Schulz, H.-J., Streit, M., & Tominski, C. (2017). Characterizing guidance in visual analytics. *IEEE Transactions on Visualization and Computer Graphics*, *23*(1), 111–120. https://doi.org/10.1109/tvcg.2016.2598468

Cheddad, A., Condell, J., Curran, K., & Mc Kevitt, P. (2010). Digital image steganography: Survey and analysis of current methods. *Signal Processing*, *90*(3), 727–752. https://doi.org/10.1016/j.sigpro.2009.08.010

Chen, Q., Sun, F., Xu, X., Chen, Z., Wang, J., & Cao, N. (2022). VizLinter: A linter and fixer framework for data visualization. *IEEE Transactions on Visualization and Computer Graphics*, *28*(1), 206–216. https://doi.org/10.1109/tvcg.2021.3114804

Chen, Y., Yang, J., & Ribarsky, W. (2009). Toward effective insight management in visual analytics systems. *Proceedings of IEEE Pacific Visualization Symposium* (pp. 49–56). https://doi.org/10.1109/pacificvis.2009.4906837

Cho, K., van Merrïenboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). *Learning phrase representations using RNN encoder–decoder for statistical machine translation* [Paper presentation]. Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (pp. 1724–1734). https://doi.org/10.3115/v1/D14-1179

Cui, Z., Badam, S. K., Yalçin, M. A., & Elmqvist, N. (2019). DataSite: Proactive visual data exploration with computation of insight-based recommendations. *Information Visualization*, *18*(2), 251–267. https://doi.org/10.1177/1473871618806555

Demiralp, C., Haas, P. J., Parthasarathy, S., & Pedapati, T. (2017). *Foresight: Rapid data exploration through guideposts* [Paper presentation]. Workshop on Data Systems for Interactive Analysis (DSIA) at IEEE VIS 2017.

Demiralp, C., Haas, P. J., Parthasarathy, S., & Pedapati, T. (2017). Foresight: Recommending visual insights. *Proceedings of the VLDB Endowment*, *10*(12), 1937–1940. https://doi.org/10.14778/3137765.3137813

Demiralp, C., Scheidegger, C. E., Kindlmann, G. L., Laidlaw, D. H., & Heer, J. (2014). Visual embedding: A model for visualization. *IEEE Computer Graphics and Applications*, *34*(1), 10–15. https://doi.org/10.1109/mcg.2014.18

Devlin, J., Chang, M., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of deep bidirectional transformers for language understanding* [Paper presentation]. Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (pp. 4171–4186). https://doi.org/10.18653/V1/N19-1423

Ding, R., Han, S., Xu, Y., Zhang, H., & Zhang, D. (2019). *QuickInsights: Quick and automatic discovery of insights from multi-dimensional data* [Paper presentation]. Proceedings of International Conference on Management of Data (pp. 317–332). https://doi.org/10.1145/3299869.3314037

Espadoto, M., Martins, R. M., Kerren, A., Hirata, N. S., & Telea, A. C. (2021). Toward a quantitative survey of dimension reduction

techniques. *IEEE Transactions on Visualization and Computer Graphics*, *27*(3), 2153–2173. https://doi.org/10.1109/TVCG.2019.2944182

Gu, J., Wang, Z., Kuen, J., Ma, L., Shahroudy, A., Shuai, B., Liu, T., Wang, X., Wang, G., Cai, J., & Chen, T. (2018). Recent advances in convolutional neural networks. *Pattern Recognition*, *77*(C), 354–377. https://doi.org/10.1016/j.patcog.2017.10.013

Hu, K., Bakker, M. A., Li, S., Kraska, T., & Hidalgo, C. (2019). *VizML: A machine learning approach to visualization recommendation* [Paper presentation]. Proceedings of ACM Conference on Human Factors in Computing Systems (pp. 1–12). https://doi.org/10.1145/3290605.3300358

Hullman, J., Drucker, S., Riche, N. H., Lee, B., Fisher, D., & Adar, E. (2013). A deeper understanding of sequence in narrative visualization. *IEEE Transactions on Visualization and Computer Graphics*, *19*(12), 2406–2415. https://doi.org/10.1109/tvcg.2013.119

Jankun-Kelly, T., Ma, K.-L., & Gertz, M. (2007). A model and framework for visualization exploration. *IEEE Transactions on Visualization and Computer Graphics*, *13*(2), 357–369. https://doi.org/10.1109/tvcg.2007.28

Jebb, A. T., Parrigon, S., & Woo, S. E. (2017). Exploratory data analysis as a foundation of inductive research. *Human Resource Management Review*, *27*(2), 265–276. https://doi.org/10.1016/j.hrmr.2016.08.003

Jiang, Q., Sun, G., Dong, Y., & Liang, R. (2021). DT2VIS: A focus + context answer generation system to facilitate visual exploration of tabular data. *IEEE Computer Graphics and Applications*, *41*(5), 45–56. https://doi.org/10.1109/mcg.2021.3097326

Jiang, Q., Sun, G., Li, T., Tang, J., Xia, W., Zhu, S., & Liang, R. (2024). Qutaber: Task-based exploratory data analysis with enriched context awareness. *Journal of Visualization*, *27*(3), 503–520. https://doi.org/10.1007/s12650-024-00975-1

Joglekar, M., Garcia-Molina, H., & Parameswaran, A. (2019). Interactive data exploration with smart drill-down. *IEEE Transactions on Knowledge and Data Engineering*, *31*(1), 46–60. https://doi.org/10.1109/TKDE.2017.2685998

Keim, D., Kohlhammer, J., Ellis, G., & Mansmann, F. (2010). *Mastering the information age solving problems with visual analytics*. Eurographics Association. https://doi.org/10.2312/14803

Kim, Y., Wongsuphasawat, K., Hullman, J., & Heer, J. (2017). *GraphScape: A model for automated reasoning about visualization similarity and sequencing* [Paper presentation]. Proceedings of ACM Conference on Human Factors in Computing Systems (pp. 2628–2638). https://doi.org/10.1145/3025453.3025866

Kusner, M. J., Paige, B., & Hernández-Lobato, J. M. (2017). Grammar variational autoencoder. *Proceedings of International Conference on Machine Learning* (pp. 1945–1954).

Law, P.-M., Basole, R. C., & Wu, Y. (2018). Duet: Helping data analysis novices conduct pairwise comparisons by minimal specification. *IEEE Transactions on Visualization and Computer Graphics*, *25*(1), 427–437. https://doi.org/10.1109/tvcg.2018.2864526

Law, P.-M., Das, S., & Basole, R. C. (2019). *Comparing apples and oranges: Taxonomy and design of pairwise comparisons within tabular data* [Paper presentation]. Proceedings of ACM Conference on Human Factors in Computing Systems (pp. 1–12). https://doi.org/10.1145/3290605.3300409

Li, H., Wang, Y., Wu, A., Wei, H., & Qu, H. (2022). *Structure-aware visualization retrieval* [Paper presentation]. Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems. https://doi.org/10.1145/3491102.3502048

Li, J. K., Xu, S., Ye, Y., & Ma, K.-L. (2020). Resolving conflicting insights in asynchronous collaborative visual analysis. *Computer Graphics Forum*, *39*(3), 497–509. https://doi.org/10.1111/cgf.13997

Li, Q., Lin, H., Tang, C. F., Wei, X., Peng, Z., Ma, X., & Chen, T. (2021). *Exploring the "double-edged sword" effect of auto-insight recommendation in exploratory data analysis* [Paper presentation]. IUI Workshops.

Lingam, S. N., Franssen, M., Petermeijer, S. M., & Martens, M. (2024). Challenges and future directions for human-drone interaction research: An expert perspective. *International Journal of Human–Computer Interaction*, 1–17. https://doi.org/10.1080/10447318.2024.2400756

Lu, J., Chen, W., Ye, H., Wang, J., Mei, H., Gu, Y., Wu, Y., Zhang, X. L., & Ma, K.-L. (2021). *Automatic generation of unit visualization-based scrollytelling for impromptu data facts delivery* [Paper presentation]. 2021 IEEE 14th Pacific Visualization Symposium (PacificVis) (pp. 21–30). https://doi.org/10.1109/PacificVis52677.2021.00011

Ma, K.-L. (1999). Image graphs – A novel approach to visual data exploration. *Proceedings of IEEE Conference on Visualization* (pp. 81–88). https://doi.org/10.1109/visual.1999.809871

Mackinlay, J., Hanrahan, P., & Stolte, C. (2007). Show me: Automatic presentation for visual analysis. *IEEE Transactions on Visualization and Computer Graphics*, *13*(6), 1137–1144. https://doi.org/10.1109/tvcg.2007.70594

Matplotlib. (2022). https://matplotlib.org/.

McInnes, L., Healy, J., Saul, N., & Großberger, L. (2018). Umap: Uniform manifold approximation and projection. *Journal of Open Source Software*, *3*(29), 861. https://doi.org/10.21105/joss.00861

Microsoft Power BI. (2022). https://powerbi.microsoft.com/en-us/

Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. *Advances in Neural Information Processing Systems* (Vol. 26, pp. 3111–3119). https://doi.org/10.1007/0-387-28981-x

Moritz, D., Wang, C., Nelson, G. L., Lin, H., Smith, A. M., Howe, B., & Heer, J. (2018). Formalizing visualization design knowledge as constraints: Actionable and extensible models in Draco. *IEEE Transactions on Visualization and Computer Graphics*, *25*(1), 438–448. https://doi.org/10.1109/tvcg.2018.2865240

Nowak, A., Piccinno, F., & Altun, Y. (2024). Multimodal chart retrieval: A comparison of text, table and image based approaches. *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* (pp. 5488–5505). https://doi.org/10.18653/v1/2024.naacllong.307

Pérez-Messina, I., Ceneda, D., El-Assady, M., Miksch, S., & Sperrle, F. (2022). A typology of guidance tasks in mixed-initiative visual analytics environments. *Computer Graphics Forum*, *41*(3), 465–476. https://doi.org/10.1111/cgf.14555

Sarvghad, A., Tory, M., & Mahyar, N. (2017). Visualizing dimension coverage to support exploratory analysis. *IEEE Transactions on Visualization and Computer Graphics*, *23*(1), 21–30. https://doi.org/10.1109/TVCG.2016.2598466

Satyanarayan, A., Moritz, D., Wongsuphasawat, K., & Heer, J. (2017). Vega-Lite: A grammar of interactive graphics. *IEEE Transactions on Visualization and Computer Graphics*, *23*(1), 341–350. https://doi.org/10.1109/tvcg.2016.2599030

Shi, D., Xu, X., Sun, F., Shi, Y., & Cao, N. (2021). Calliope: Automatic visual data story generation from a spreadsheet. *IEEE Transactions on Visualization and Computer Graphics*, *27*(2), 453–463. https://doi.org/10.1109/tvcg.2020.3030403

Sperrle, F., Ceneda, D., & El-Assady, M. (2023). Lotse: A practical framework for guidance in visual analytics. *IEEE Transactions on Visualization and Computer Graphics*, *29*(1), 1124–1134. https://doi.org/10.1109/tvcg.2022.3209393

Srinivasan, A., Drucker, S. M., Endert, A., & Stasko, J. (2018). Augmenting visualizations with interactive data facts to facilitate interpretation and communication. *IEEE Transactions on Visualization and Computer Graphics*, *25*(1), 672–681. https://doi.org/10.1109/tvcg.2018.2865145

Stoiber, C., Ceneda, D., Wagner, M., Schetinger, V., Gschwandtner, T., Streit, M., Miksch, S., & Aigner, W. (2022). Perspectives of visualization onboarding and guidance in VA. *Visual Informatics*, *6*(1), 68–83. https://doi.org/10.1016/j.visinf.2022.02.005

Sun, G., Zhu, S., Jiang, Q., Xia, W., & Liang, R. (2022). EvoSets: Tracking the sensitivity of dimensionality reduction results across subspaces. *IEEE Transactions on Big Data*, *8*(6), 1566–1579. https://doi.org/10.1109/TBDATA.2021.3079200

Tableau. (2022). https://www.tableau.com/

Tang, B., Han, S., Yiu, M. L., Ding, R., & Zhang, D. (2017). *Extracting top-k insights from multi-dimensional data* [Paper presentation]. Proceedings of ACM International Conference on Management of Data (pp. 1509–1524). https://doi.org/10.1145/3035918.3035922

Tukey, W. J. (1977). *Exploratory data analysis.*
Valero-Mora, P. M. (2010). ggplot2: Elegant graphics for data analysis. *Journal of Statistical Software*, *35*(Book Review 1), 1–3. https://doi.org/10.18637/jss.v035.b01
Van der Maaten, L., & Hinton, G. (2008). Visualizing data using t-SNE. *Journal of Machine Learning Research*, *9*(11), 2579–2605.
Vartak, M., Huang, S., Siddiqui, T., Madden, S., & Parameswaran, A. (2017). Towards visualization recommendation systems. *ACM SIGMOD Record*, *45*(4), 34–39. https://doi.org/10.1145/3092931.3092937
Vartak, M., Rahman, S., Madden, S., Parameswaran, A., & Polyzotis, N. (2015). SeeDB: Efficient data-driven visualization recommendations to support visual analytics. *Proceedings of the VLDB Endowment*, *8*(13), 2182–2193. https://doi.org/10.14778/2831360.2831371
Wang, Y., Sun, Z., Zhang, H., Cui, W., Xu, K., Ma, X., & Zhang, D. (2020). DataShot: Automatic generation of fact sheets from tabular data. *IEEE Transactions on Visualization and Computer Graphics*, *26*(1), 895–905. https://doi.org/10.1109/tvcg.2019.2934398
Wongsuphasawat, K., Liu, Y., Heer, J. (2019). *Goals, process, and challenges of exploratory data analysis: An interview study*. arXiv preprint arXiv:1911.00568. https://doi.org/10.48550/arXiv.1911.00568
Wongsuphasawat, K., Moritz, D., Anand, A., Mackinlay, J., Howe, B., & Heer, J. (2016). Voyager: Exploratory analysis via faceted browsing of visualization recommendations. *IEEE Transactions on Visualization and Computer Graphics*, *22*(1), 649–658. https://doi.org/10.1109/tvcg.2015.2467191
Wongsuphasawat, K., Qu, Z., Moritz, D., Chang, R., Ouk, F., Anand, A., Mackinlay, J., Howe, B., & Heer, J. (2017). *Voyager 2: Augmenting visual analysis with partial view specifications* [Paper presentation]. Proceedings of ACM Conference on Human Factors in Computing Systems (pp. 2648–2659). https://doi.org/10.1145/3025453.3025768
Wu, A., Wang, Y., Shu, X., Moritz, D., Cui, W., Zhang, H., Zhang, D., & Qu, H. (2022). AI4VIS: Survey on artificial intelligence approaches for data visualization. *IEEE Transactions on Visualization and Computer Graphics*, *28*(12), 5049–5070. https://doi.org/10.1109/TVCG.2021.3099002
Xie, P., Tao, W., Li, J., Huang, W., & Chen, S. (2021). Exploring multi-dimensional data via subset embedding. *Computer Graphics Forum*, *40*(3), 75–86. https://doi.org/10.1111/cgf.14290
Xu, S., Bryan, C., Li, J. K., Zhao, J., & Ma, K.-L. (2018). Chart constellations: Effective chart summarization for collaborative and multi-user analyses. *Computer Graphics Forum*, *37*(3), 75–86. https://doi.org/10.1111/cgf.13402
Zeng, Z., Moh, P., Du, F., Hoffswell, J., Lee, T. Y., Malik, S., Koh, E., & Battle, L. (2022). An evaluation-focused framework for visualization recommendation algorithms. *IEEE Transactions on Visualization and Computer Graphics*, *28*(1), 346–356. https://doi.org/10.1109/TVCG.2021.3114814
Zhang, G., Sun, G., Sun, Z., Tang, J., Jiang, L., & Liang, R. (2024). Towards better utilization of haptic interaction in visualization: Design space and knob prototype. *International Journal of Human–Computer Interaction*, 1–22. https://doi.org/10.1080/10447318.2024.2360772
Zhang, S., Li, H., Qu, H., & Wang, Y. (2024). Adavis: Adaptive and explainable visualization recommendation for tabular data. *IEEE Transactions on Visualization and Computer Graphics*, *30*(9), 5923–5938. https://doi.org/10.1109/TVCG.2023.3316469
Zhao, J., Fan, M., & Feng, M. (2022). ChartSeer: Interactive steering exploratory visual analysis with machine intelligence. *IEEE Transactions on Visualization and Computer Graphics*, *28*(3), 1500–1513. https://doi.org/10.1109/tvcg.2020.3018724
Zhu, S., Sun, G., Jiang, Q., Zha, M., & Liang, R. (2020). A survey on automatic infographics and visualization recommendations. *Visual Informatics*, *4*(3), 24–40. https://doi.org/10.1016/j.visinf.2020.07.002

About the authors

Qi Jiang is working toward the PhD degree in computer science and technology at Zhejiang University of Technology. He received the BE degree in electrical engineering and automation from the Zhejiang University of Technology in 2019. His research interests include visual analytics and intelligent visualization.

Guodao Sun is a professor at Zhejiang University of Technology. He received the BSc in computer science and technology, and PhD degree in control science and engineering both from Zhejiang University of Technology. His research interest is information visualization and he is the corresponding author.

Yue Dong is currently working toward the ME degree in electronic information from the Zhejiang University of Technology, Hangzhou, China. She received her BE degree in software engineering from the Zhejiang University of Technology, in 2020. Her main research interests are information visualization and visual analytics.

Lvhan Pan is currently working toward the ME degree in computer science and technology from the Zhejiang University of Technology, Hangzhou, China. He received the BSc in Software Engineering, Zhejiang University of Technology. His main research interests are information visualization, visual analytics.

Baofeng Chang received PhD degree in Computer Science and Technology at the College of Computer Science and Technology, Zhejiang University of Technology, Hangzhou, China. He received BSc degree in Automation in the College of Information Engineering, Zhejiang University of Technology. His main research interests are information visualization and visual analytics.

Li Jiang is an associate professor at the College of Computer Science and Technology, Zhejiang University of Technology, Hangzhou, China. She received her PhD degree in Computer Science and Technology from Zhejiang University of Technology. Her main research interests are urban visualization and visual analytics of social media.

Haoran Liang is an associate researcher at the College of Computer Science and Technology, Zhejiang University of Technology. He received the BSc in computer science and technology, and PhD degree in control science and engineering both from Zhejiang University of Technology. His research interests include computer vision and visualization.

Ronghua Liang received the PhD in computer science from Zhejiang University. He is currently a professor of Computer Science and Dean of College of Computer Science and Technology, Zhejiang University of Technology, China. His research interests include visual analytics and computer vision.